Provably Good Prim–Dijkstra Revisited: New Theory and a Practical Algorithm for a Classical VLSI Routing Problem with LLMs

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Abstract

Large language models may make precise but dormant algorithmic problems practical to revisit, and may expose new paths toward fundamental ones. We demonstrate this possibility through Prim–Dijkstra routing, a classic VLSI problem whose terminal-only Manhattan complexity remained open despite decades of practical work. We prove weak NP-completeness, derive a continuous cost–radius tradeoff with a balanced $(2, 2)$ guarantee, and build HP-RCRST, a height-partition-based multi-mode solver. On 28 development instances, its stronger modes Pareto-dominate the published-method union on 23 and tie on five. The case shows how conflicting conjectures, counterexamples, formal checks, and implementation can reopen neglected questions. Code and reproducibility materials are available at https://github.com/CODA-Team/hp-rcrst.

Keywords

routing trees, Prim–Dijkstra, bicriteria approximation, VLSI physical design, large language models

1 Introduction

Some questions wait in the literature. Others are put to work while they wait. Prim–Dijkstra routing has lived in both worlds for more than three decades.

A routing tree may be short when viewed from the plane and long when viewed from its source. A minimum spanning tree minimizes total edge length but can send a source–sink path through a long detour; a shortest-path tree minimizes those paths but may pay for many direct connections. Prim and Dijkstra occupy the two ends of this tension. Both are among the first greedy algorithms a student encounters. Putting them on the same dial creates a problem that can be stated in a few lines and has remained part of physical-design research ever since. Figure 1 shows the disagreement on a small instance and the kind of tree sought between the two endpoints.

The early performance-driven routing literature did not treat this tension as mere engineering folklore. In 1992, Cong et al. formulated bounded-radius routing trees and developed algorithms that were, in the language of their title, *provably good* [6]. Their paper also left the complexity of the minimum-cost bounded-radius spanning tree in the Manhattan plane as an open question. Three years later, Alpert et al. joined Prim and Dijkstra in a single parameterized construction [2]. The resulting PD family became a durable way to navigate the wirelength–delay tradeoff.

The line did not end. Shallow-light trees gave a general-graph guarantee by changing the form of the question [18]. Allowing Steiner vertices changes the attainable lightness, and geometric variants continue to refine that distinction [8, 19, 26]. SALT carried provable shallow-light reasoning into rectilinear Steiner routing [4]. In 2018, *Prim–Dijkstra Revisited* showed that the elementary PD construction still admitted substantial practical improvement [1]. Recent multi-source variants moved the objective toward cost–skew tradeoffs [16]. Across these changes, the same two quantities kept returning: how much tree is built, and how far the tree makes a signal travel.

Yet the original terminal-only Manhattan problem remained between two communities. For EDA it was familiar enough that a heuristic could be used without first settling its exact complexity. For theoretical computer science it looked application-specific beside the general theory of shallow-light trees. The result was an unusual asymmetry: the construction was repeatedly improved, but the most basic complexity question and a direct global-radius approximation theory were not brought together with a modern implementation.

No contemporary routing flow is waiting for this classification. Its value now lies in the class of problems it represents: precise, locally checkable questions that slipped out of active view as communities and tools moved on. Reopening such a question once required enough literature recovery, proof work, and software reconstruction to make the attempt difficult to justify. LLMs change that threshold. More of these questions can now be revisited, and some may reveal theoretical or practical opportunities that were never visible when they were set aside. The same parallel cycle of conjecture, attack, formalization, and experiment may also open new routes into fundamental problems.

Recent language-model research now touches two neighboring forms of inquiry. In mathematics, OpenAI has released a GPT-generated proof of the cycle double cover conjecture together with the prompt that produced it [22], while formal-mathematics agents are beginning to address open-ended research questions [27]. AlphaGeometry couples learned guidance with symbolic deduction, while LeanDojo couples language models to a proof-assistant environment [28, 29]. Evaluator-guided systems such as FunSearch and AlphaEvolve treat programs and algorithms as objects of discovery [11, 23]. In EDA, multi-agent self-evolution modifies, evaluates, and retains improvements to the design tools themselves [30].

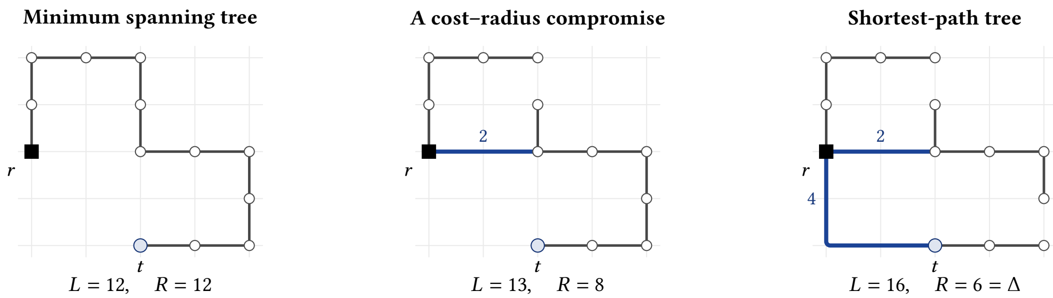

Figure 1: The Prim–Dijkstra problem between its two textbook endpoints. The same 13 terminals are used in all three panels. Unit chain edges form an MST with $(L, R) = (12, 12)$. One Manhattan shortcut gives the middle tree $(13, 8)$; a second produces an SPT $(16, 6)$. Labeled heavy paths depict complete-metric edges with the shown Manhattan lengths; an unlabeled bend is not a Steiner vertex. The cost–radius problem asks which middle-ground trees satisfy prescribed bounds on both quantities.

Mathematics and EDA approach the new machinery from opposite directions: one through conjectures and proofs, the other through executable tools and objective evaluators. The old Prim–Dijkstra question sits between them. It is a theorem question embedded in a practical routing construction. The construction has been repeatedly used and improved; the complexity question remains where it was left.

That position makes the problem unusually suitable for a model-assisted revisit. Its definition is short; candidate trees and counterexamples are inexpensive to check; mathematical claims can be tested against executable instances; and the implementation has a small, inspectable engineering surface. Proof validity, problem choice, and final research judgment remain the author's responsibility.

For this revisit, we organized a set of deliberately conflicting investigations. Some were asked to prove polynomial-time solvability without access to prior answers; another was asked to establish hardness; others pursued approximation and implementation with progressively broader information. Failed claims were preserved as counterexamples, and the empirical objective was frozen before the final solver was developed. The author selected the problem and experimental conditions, reconstructed the arguments, performed the literature due diligence, and verified the claims reported here.

Our contributions are as follows:

- We settle the complexity question with an explicit integer reduction. It proves NP-completeness for the rooted, terminal-only Manhattan cost–radius problem. The reduction is weak and numerical. To the best of our knowledge, this answers the question left open in 1992.
- We give a bottom-up height partition of a rooted MST. For a continuous parameter H, it bounds global radius by $\Delta + H$ and excess length through H; the balanced choice gives a $(2, 2)$ cost–radius guarantee. The parameter H continues the sequence of knobs that has shaped this literature.
- We implement the theory in HP-RCRST, a self-contained deterministic solver with one certified mode and three practical runtime–quality modes. On the 28 development instances, its stronger modes Pareto-dominate the published-method union on 23 cases and tie it on the other five; the full mode-by-mode comparison appears in Section 6.
- We release the five task specifications and information boundaries used in the study, together with the final solver and checker, selected benchmarks, baseline adaptations, frozen result bundles, and the completed sorry-free Lean modules. The public artifact is a reproducibility subset of the research record, not a transcript of every model interaction.

2 A Classical Cost–Radius Problem

2.1 Model

Let $V \subset \mathbb{Z}^2$ be a set of distinct terminals and let $r \in V$ be the source. Edges belong to the complete terminal graph and have Manhattan length $d(u, v)$. For a spanning tree T, let

$$L(T) = \sum_{e \in T} d(e) \qquad \text{and} \qquad R(T) = \max_{v \in V} d_T(r, v).$$

The decision problem asks whether a tree exists with $L(T) \leq B$ and $R(T) \leq D$. We write M for the exact terminal MST length and $\Delta = \max_v d(r, v)$. Every feasible comparison tree satisfies $L(T) \geq M$ and $R(T) \geq \Delta$. We therefore summarize both objectives by the normalized performance vector

$$\boldsymbol{\rho}(T) := \left(\frac{L(T)}{M}, \frac{R(T)}{\Delta} \right), \qquad \boldsymbol{\rho}(T) \succeq (1, 1),$$

where vector inequalities are componentwise and smaller is better.

Example 2.1 (The running instance). For the 13 terminals in Figure 1, $M = 12$ and $\Delta = 6$. The three displayed trees have

$$\boldsymbol{\rho}(T_{\mathrm{MST}}) = (1, 2), \qquad \boldsymbol{\rho}(T_{\mathrm{mid}}) = \left(\frac{13}{12}, \frac{4}{3} \right), \qquad \boldsymbol{\rho}(T_{\mathrm{SPT}}) = \left(\frac{4}{3}, 1 \right).$$

With budgets $B = 13$ and $D = 8$, the MST violates only the radius bound, the SPT violates only the length bound, and the middle tree satisfies both. Thus neither endpoint solves even this small rectangular decision instance.

This paper keeps three boundaries explicit. Trees are terminal-only; the radius is measured from a designated source rather than by all-pairs diameter; and the same Manhattan metric determines both total cost and path length. General weighted graphs, Euclidean geometry, Steiner points, per-terminal stretch, and skew objectives are important neighbors, but they are not the same problem.

2.2 Constructions, knobs, and guarantees

The two endpoints optimize one coordinate each:

$$L(T_{\mathrm{MST}}) = M, \qquad R(T_{\mathrm{SPT}}) = \Delta.$$

Metric status. Every empirical point and every tree passed to our evaluator uses the complete terminal graph with $d(u,v) = \|u - v\|_1$. The endpoint identities and our height-partition theorem hold in any finite metric. KRY/LAST is stated more generally for a non-negatively weighted connected graph, and hence applies to this complete Manhattan graph. BRBC, PD, and PD-II were developed in the rectilinear VLSI-routing setting, while MSPD/MSS also changes the source and skew objective. When these methods enter our comparison, we retain only their terminal parent trees and recompute L and R in the present single-source Manhattan model. A common metric therefore does not mean that the original papers solve the same optimization problem.

KRY recasts the tradeoff as a light approximate shortest-path tree (LAST) [18]. For every $\alpha > 1$, it constructs a tree T_α satisfying

$$d_{T_\alpha}(r,v) \le \alpha\, d(r,v) \quad (\forall v), \qquad L(T_\alpha) \le \left(1 + \frac{2}{\alpha - 1}\right) M.$$

Consequently,

$$\boldsymbol{\rho}(T_\alpha) \preceq \left(1 + \frac{2}{\alpha - 1},\, \alpha\right).$$

The symmetric setting is $\alpha = 1 + \sqrt{2}$, giving the common factor $1 + \sqrt{2}$. This is stronger than a global-radius statement on its path coordinate: it controls every terminal relative to that terminal's own shortest distance.

The EDA constructions use their knobs differently. The bounded-radius, bounded-cost (BRBC) family starts from an MST traversal and inserts root connections when a radius threshold is crossed [6]. Prim–Dijkstra (PD) instead grows one tree with the key

$$\mathrm{key}_\lambda(u,v) = d_T(r,u) + \lambda\, d(u,v), \qquad \lambda \ge 1,$$

moving from Dijkstra-like path growth toward Prim-like short edges as λ increases [2]. PD-II uses separate parameters for construction and detour repair, then applies edge flipping [1]; MSPD/MSS further varies the source set and targets cost–skew rather than the exact rectangle studied here [16]. These parameter values are not interchangeable, and PD and PD-II do not inherit the LAST bound above.

Our parameter H returns to a direct global cost–radius statement:

$$\boldsymbol{\rho}(A_H) \preceq \left(1 + \frac{\Delta}{H},\, 1 + \frac{H}{\Delta}\right), \qquad H = \Delta \implies \boldsymbol{\rho}(A_H) \preceq (2,2).$$

Thus KRY/LAST and HP-RCRST trace the envelopes $(x-1)(y-1) = 2$ and $(x-1)(y-1) = 1$, respectively, in the normalized cost–radius plane. The second curve is tighter for global radius; it does not replace KRY's stronger per-terminal stretch theorem.

2.3 Neighboring models

The related theory follows several parallel routes. In general graphs, delay-constrained spanning trees were shown hard and studied through network heuristics, mathematical programming, and bi-objective optimization [3, 12, 24]. Those formulations permit an arbitrary input graph and, in several cases, separate cost and delay

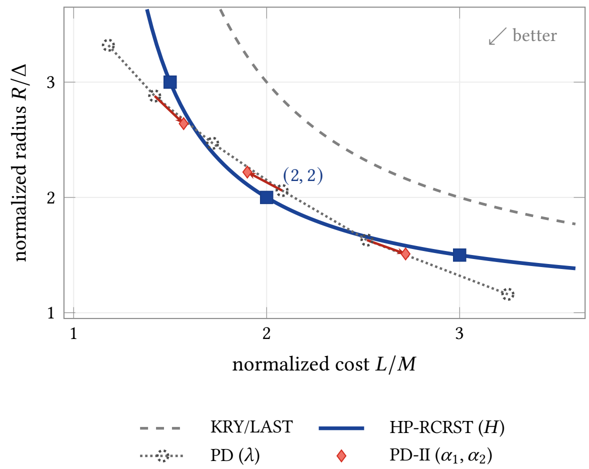

Figure 2: The knob lineage on one normalized cost–radius plane. The KRY/LAST and HP-RCRST curves are worst-case upper-bound envelopes from the displayed bounds, not empirical frontiers. Hollow points and red repair arrows show only the roles of the PD and PD-II parameters; their plotted positions are schematic and carry neither a numerical claim nor a method ranking.

weights. They therefore do not settle the complete terminal graph in which one Manhattan metric supplies both quantities.

Geometric variants also impose different path constraints. Bounded-path spanning and Steiner trees have been treated by exact and heuristic methods [21], and single-source dilation bounds compare each root path with that terminal's direct Euclidean distance [5]. Diameter and minimum-radius variants supply further hardness results [15, 25]. A per-terminal multiplicative dilation bound, an all-pairs diameter bound, and our single global additive budget D define distinct feasible sets.

A second neighboring line replaces the pair of hard bounds by a scalar cost–distance objective: total construction cost plus weighted source–sink path lengths. It begins with two-metric network design [20]; uniform variants admit progressively sharper approximations [9, 17]. Related EDA work embeds timing, congestion, buffering, or multi-net resource sharing into larger optimization systems [7, 13, 14]. These models explain the practical reach of the same cost–delay tension, but their weighted-sum objectives, Steiner vertices, routing graphs, and global design constraints are not the terminal-only Manhattan decision problem studied here. Thus the results above narrow, rather than remove, the novelty of our theorem: its role is to close that precise case and connect the answer to a particularly simple approximation.

3 Revisiting the Problem with Language Models

We used the same model—OpenAI GPT-5.6 Sol in Codex with ultra reasoning effort—under five deliberately incompatible research conditions. What changed was the premise, the information boundary, and the object that had to survive external checking:

Hardness. Starting from the problem definition alone, construct and adversarially audit an NP-completeness proof.

Exact, blind. Pursue a polynomial-time exact algorithm without literature, prior algorithms, Internet access, or permission to retreat to a hardness claim.

Exact, informed. Pursue the same conclusion with local papers and baseline code, but without access to the other investigations.

Approximation. Assume hardness, consult the literature, and seek both a provable bicriteria guarantee and a strong empirical frontier.

Synthesis. Inspect the accumulated record and build the final solver against an evaluator frozen before its implementation.

The two exact conditions provide a useful control. Despite their different information, both arrived at essentially the same architecture:

$$\begin{array}{c}\text{blind exact} \quad \| \quad \text{informed exact}\\ \Downarrow \\ \text{parent-arc master + lazy connectivity/radius cuts} \\ \Downarrow \\ \text{same gap: rounds, pivots, and rational bit growth}\end{array}$$

Polynomial-time separation of the individual cuts did not give a polynomial-time exact algorithm. A concrete scalarization witness was the three-point frontier

$$\mathcal{F} = \{(51, 20), \boxed{(42, 23)}, (34, 24)\},$$
$$(42, 23) \in \arg\min_{(L,R)\in\mathcal{F}} (L + \lambda R) \implies \lambda \le 3 \ \wedge \ \lambda \ge 8.$$

Thus the boxed Pareto point is unsupported. The blind condition found an analogous witness, and both conditions found dominated local-exchange optima. We retained the formulations as small-instance oracles and the counterexamples as boundaries; neither exact condition was reported as a proof that the problem is in P.

Example 3.1 (A counterexample that changed the proof object). The approximation condition conjectured that the safe minimum of two $\alpha = 2$ LAST-style endpoint policies always contains a factor-two tree. The claim survived exhaustive and randomized testing, but an $n = 58$ subdivided broom has

$$M = 57, \qquad \Delta = 9, \quad (L, R) = (117, 12),$$
$$\max\left\{\frac{L}{M}, \frac{R}{\Delta}\right\} = \frac{39}{19} > 2.$$

The witness redirected the proof from per-terminal stretch to the global maximum radius used by the objective. Charging whole residual components instead yields

$$R(A_H) \le \Delta + H, \qquad \big(L(A_H) - M\big)H \le M\Delta.$$

These are the height-partition inequalities of Theorem 4.2, not a repaired version of the falsified invariant.

Synthesis continued the approximation route. The final condition did not assemble all preceding solvers. Its inherited spine and its additions were:

Inherited. Classical anchors, height and centered partitions, exact objective reconstruction, the frozen evaluator, and portfolio comparison.

Added. Bounded reparenting and component exchange, cap repair, separately budgeted primary and companion search branches, deterministic portfolio selection, and size- and geometry-dependent dispatch.

Not imported. The two rational cutting-plane engines, whose uncontrolled exact-search tails conflicted with the production role.

The implementation was largely rewritten, but the line of attack continued: the CERTIFIED mode preserves the theorem-bearing construction, while the empirical modes begin from the same pool and spend additional work on frontier coverage.

Each condition maintained claim and counterexample ledgers. Candidate theorems were converted into explicit inequalities, candidate algorithms were checked on exact small instances, and negative witnesses were passed forward instead of discarded. The author selected the problem and information boundaries, reconstructed the proofs, checked the literature boundary, fixed the evaluator, and owns the final claims.

The public repository is a reproducibility subset rather than a transcript. It releases the five task specifications, final solver and checkers, selected benchmarks and baseline adaptations, frozen results, and completed Lean modules. It does not release complete conversations, private references, unfinished formalizations, or every exploratory ledger and program. Appendix E gives the exact inventory and formalization boundary.

4 New Theory

The two theoretical results address different questions. The first explains why exact optimization has resisted the sequence of simple constructions reviewed in Section 2. The second shows that hardness does not prevent a particularly simple global cost–radius guarantee.

4.1 NP-completeness via a weak reduction

THEOREM 4.1 (COMPLEXITY OF RECTILINEAR RCRST). *The rooted, terminal-only Manhattan cost–radius spanning-tree decision problem is NP-complete under polynomial-time many-one reductions. The reduction establishes weak NP-hardness; it does not establish strong NP-hardness.*

Membership in NP follows from an $n-1$-edge certificate. Exact binary arithmetic gives every edge length and total cost, while one rooted traversal computes all source distances and the radius.

For hardness, reduce from positive-integer PARTITION. Given $a_1, \dots, a_n$, let

$$S = \sum_{i=1}^{n} a_i, \qquad h = 10S, \qquad t = 4S, \qquad X = 2hn + t.$$

Create axis terminals and upper terminals

$$s_i = (2hi, 0) \quad (0 \le i \le n), \qquad u_i = (2hi - 3a_i, a_i) \quad (1 \le i \le n),$$

together with $z = (X, 0)$; the root is s_0. Set

$$B = X + 3S, \qquad D = X + S.$$

The strict left-to-right order is $s_0, u_1, s_1, \dots, u_n, s_n, z$, because every long interblock gap is at least $17S$.

Algorithm 1 HeightPartition(T_M, r, H)

Require: Rooted MST T_M, root r, threshold $H > 0$
Ensure: A terminal spanning tree A_H

$A_H \leftarrow T_M$; $h_v \leftarrow 0$ for every $v \in V$
for all $v \in V \setminus \{r\}$ in postorder **do**
  $u \leftarrow \mathrm{parent}_{T_M}(v)$; $e \leftarrow d(u, v)$
  if $e + h_v \le H$ **then**
    $h_u \leftarrow \max\{h_u, e + h_v\}$
  else
    $A_H \leftarrow A_H - \{(u, v)\} + \{(r, v)\}$
  end if
end for
return A_H

If a subset I has sum $A = S/2$, each selected block uses the path $s_{i-1}u_i s_i$, while each unselected block uses $s_{i-1}s_i$ and attaches u_i as a leaf. The resulting tree has

$$L = X + 4S - 2A = B, \qquad d_T(s_0, z) = X + 2A = D,$$

and the off-path upper terminals also lie within radius D.

The reverse implication must rule out every edge of the complete Manhattan graph, not only the displayed local edges. Let P be the s_0-to-z path of an arbitrary feasible tree and put

$$F = L(T) + d_T(s_0, z) \le B + D = 2X + 4S. \tag{1}$$

Across every vertical gap in the terminal order, connectivity contributes one tree crossing and P contributes one path crossing. Their horizontal contribution already totals $2X$. A third crossing of a nonfinal interblock gap costs more than the remaining $4S$; a third crossing of the final $4S$ gap leaves no room for the strictly positive vertical cost. Every interblock prefix cut therefore has one tree crossing, which is also the unique path crossing. Nested-cut uniqueness forbids a bridge from skipping a block, and edge counting forces every internal edge $u_i s_i$.

Let W be the total bridge length and let A now denote the sum of items whose internal edge lies on P. The forced topology gives

$$L(T) = W + 4S, \qquad d_T(s_0, z) = W + 4A. \tag{2}$$

Horizontal and vertical telescoping along P, including internal edges traversed in either direction, yields the bridge inequality

$$W + 2A \ge X. \tag{3}$$

Equations (2)–(3) and $L(T) \le B$ imply $A \ge S/2$, while (2)–(3) and $d_T(s_0, z) \le D$ imply $A \le S/2$. Thus $A = S/2$, recovering a partition. Appendix A gives the complete cut and bit-complexity arguments.

4.2 A bottom-up height partition

Root an MST T_M at r. For a residual component whose top vertex is v, let h_v be its greatest retained tree distance from v. Algorithm 1 keeps the parent edge of v exactly when doing so preserves residual height at most H. Otherwise it cuts that edge and reconnects v directly to the root.

Theorem 4.2 (Global cost–radius tradeoff). *For every real $H > 0$, Algorithm 1 returns a terminal tree A_H satisfying*

$$R(A_H) \le \Delta + H, \qquad \big(L(A_H) - M\big)H \le M\Delta. \tag{4}$$

It runs in linear time after the MST has been rooted.

Proof sketch. Every residual component has height at most H. The root component therefore has radius at most H; every other component first uses one direct root edge of length at most Δ, proving the radius bound.

Consider a cut child v, its deleted parent edge of weight e_v, and the total weight W_v retained in its residual component. MST cut optimality gives

$$0 \le d(r, v) - e_v \le \Delta.$$

The failed keep test and $W_v \ge h_v$ give $e_v + W_v > H$. Charge the shortcut's net addition to these edges:

$$\big(d(r, v) - e_v\big)H \le \Delta(e_v + W_v). \tag{5}$$

Once a residual is cut, none of its edges enters an ancestor residual, so the charged edge sets are disjoint and have total weight at most M. Summing (5) proves the length inequality. Each residual component is reconnected exactly once, so the result is a tree. The full argument appears in Appendix B. □

Equivalently,

$$\boldsymbol{\rho}(A_H) \preceq \left(1 + \frac{\Delta}{H}, 1 + \frac{H}{\Delta}\right), \qquad \boxed{\boldsymbol{\rho}(A_\Delta) \preceq (2, 2)}. \tag{6}$$

The integer implementation samples the continuous envelope in (6). The balanced choice $H = \Delta$ gives a single tree whose cost and radius are each within two of their independent lower bounds.

4.3 The factor-two boundary

Theorem 4.2 uses only metricity and MST optimality. That level of generality cannot yield a universal common factor below two for a single returned tree: for every $c < 2$, an explicit finite metric admits no tree T with both $L(T) \le cM$ and $R(T) \le c\Delta$. Appendix B gives the trunk-and-arms construction.

This boundary is deliberately narrow. The construction is not embedded in the Manhattan plane, and it does not rule out a bounded portfolio whose different members cover different comparison trees. A below-two planar L_1 guarantee and a below-two approximate-Pareto portfolio theorem remain open.

5 HP-RCRST: A Certified Core and Empirical Frontiers

The theory suggests a separation that is useful in practice. One small construction family can carry a universal certificate; a larger search can then improve the measured frontier without being asked to justify that certificate. HP-RCRST makes this separation explicit. It accepts a rooted terminal set and returns a bounded portfolio of parent arrays, rather than a single tree for one preselected scalarization.

Its algorithmic spine comes from the preceding approximation investigation: classical shallow-light anchors (trees returned directly by named constructions), the height-partition and centered families, exact remeasurement, and a portfolio objective. The final implementation rewrites that spine and extends it with bounded reparenting, component exchange, cap repair, and size-dependent dispatch. Its independent search branches each have their own seed

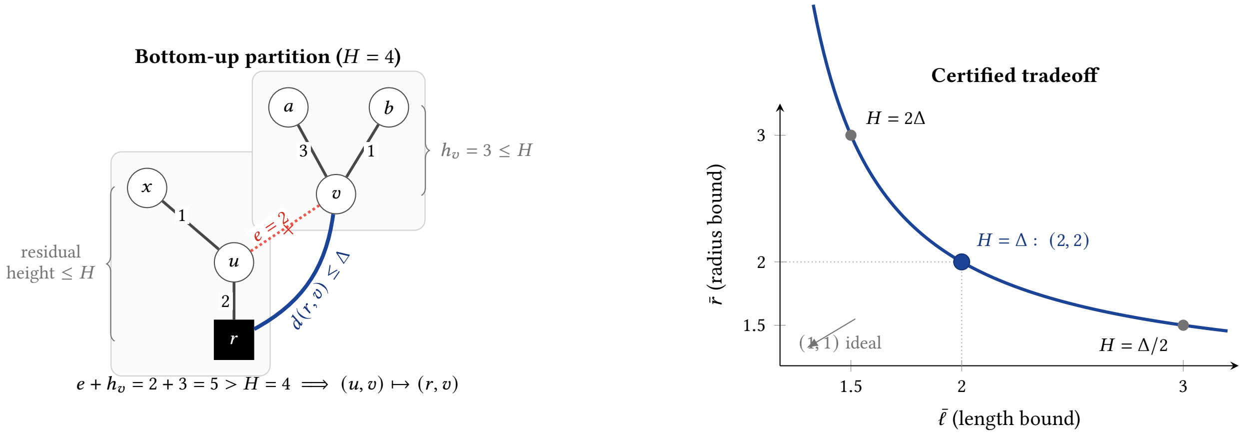

Figure 3: Bottom-up height partition and its certified tradeoff. The dotted red edge fails the residual-height test and is replaced by the blue root edge (left). The curve is the coordinatewise upper-bound envelope in (6), not an empirical Pareto frontier (right). Color supplements line weight and style; the construction remains legible in grayscale.

rule, operators, and finite budget. The earlier exact cutting-plane engines are not part of the production solver.

5.1 One exact boundary for every candidate

A candidate is represented by one parent for each nonroot terminal. Before a candidate may enter a portfolio, a common validation path checks that the parent relation is connected and acyclic, recomputes every Manhattan edge and root distance, and obtains L and R from the reconstructed tree. Incremental move formulas are used to propose trees, never to certify their reported objectives. Metric duplicates and dominated points are then removed with exact integer comparisons.

This boundary serves two purposes. It prevents a fast local update from silently changing the problem being measured, and it lets heterogeneous constructions be combined without trusting their internal bookkeeping. Products used by certificates and ratio comparisons are evaluated with unbounded integers; the supported signed-coordinate domain and overflow argument appear in Appendix C.

Five implementation terms. An *anchor* is a tree built directly by a named construction, such as MST, star, or LAST. A *seed* is a validated candidate handed to a local improvement operator. A *search branch* is one choice of seed rule, operators, and finite budget. In the centered fixed-MST family, each component keeps its MST edges and a nonroot component attaches directly from the root to one chosen center. The primary branch enables that family; the companion branch disables it and keeps a separate budget. Their validated outputs are united only after both branches finish. The nondominated result is a *frontier*; the returned *portfolio* contains at most k selected trees from that frontier.

Algorithms 2 and 3 separate the shared theorem-derived core from mode-specific work. The source contains more constructions than the pseudocode names individually, but every one enters through the same validate–nondominate boundary. The exact schedules and cutoffs are recorded in Appendix C and in the released artifact.

Algorithm 2 CERTIFIEDCORE(V, r)

Require: Terminals V, root r

$T_M \leftarrow$ DENSEPRIM(V); $\Delta \leftarrow \max_{v \in V} d(r, v)$
$A_\Delta \leftarrow$ HEIGHTPARTITION(T_M, r, Δ) ▹ Theorem 4.2
$C \leftarrow \{T_M, \text{STAR}(V, r), A_\Delta\}$
return $(C, T_M, \Delta, A_\Delta)$

5.2 The certified portfolio

The CERTIFIED mode constructs a canonical dense-Prim MST, the star, and height partitions at five logarithmically spaced thresholds. The $H = \Delta$ member is essential. If another returned tree Pareto-dominates it, the dominator may carry the certificate; portfolio truncation is otherwise forbidden to remove the witness. With $k = 1$, the exact $H = \Delta$ parent array is returned.

The dense MST costs $O(n^2)$ time and $O(n)$ auxiliary memory, and each postorder partition is linear. Thus the complete certified portfolio remains deterministic and polynomial:

$$\exists T \in \mathcal{P}_{\text{cert}} : \qquad L(T) \le 2M, \qquad R(T) \le 2\Delta. \tag{7}$$

Immediately before serialization, HP-RCRST independently rebuilds the MST and the balanced partition and rechecks both inequalities in (7).

5.3 Exact small instances and bounded improvement

The empirical modes begin with several deliberately different candidate families. Classical star, MST, PD, BRBC, and KRY/LAST constructions cover recognizable endpoint and shallow-light behaviors. Radial-label, cap-repair, and critical-height constructions add topologies not obtained by a single Prim–Dijkstra parameter sweep. These anchors are useful both as returned points and as starting trees for later moves.

Two exact mechanisms are retained only on bounded domains. The first is EXACTTREEFRONTIER for $n \le 8$. A Prüfer code is a sequence of $n - 2$ terminal indices, with repetition allowed. It is

Algorithm 3 HP-RCRST(V, r, μ, k): mode extension and selection

Require: Terminals V, root r, mode μ, portfolio cap k
$(C, T_M, \Delta, A_\Delta) \leftarrow \textsc{CertifiedCore}(V, r)$
if $\mu = \textsc{certified}$ **then**
  $C \leftarrow C \cup \{\textsc{HeightPartition}(T_M, r, H) : H \in \{\Delta/4, \Delta/2, 2\Delta, 4\Delta\}\}$
else
  $C_0 \leftarrow C \cup \textsc{ClassicalAnchors}(V, r, \mu)$
  for all independently budgeted search branches B enabled by μ **do**
    $C_B \leftarrow C_0$
    if $\textsc{CenteredEnabled}(B)$ **then**
      $C_B \leftarrow C_B \cup \textsc{CenteredPartitions}(T_M, r, \mu)$
    end if
    if $\mu \in \{\textsc{balanced}, \textsc{quality}\}$ and $|V| \leq 8$ **then**
      $C_B \leftarrow C_B \cup \textsc{ExactTreeFrontier}(V, r)$
    end if
    $\mathcal{S} \leftarrow \textsc{SelectSeeds}(C_B, B)$
    $C_B \leftarrow C_B \cup \textsc{Improve}(\mathcal{S}, B)$
    $C \leftarrow C \cup \textsc{Nondominate}(\textsc{Validate}(C_B))$
  end for
end if
$\mathcal{F} \leftarrow \textsc{Nondominate}(\textsc{Validate}(C))$
if $\mu = \textsc{certified}$ **then**
  return $\textsc{CertifiedSelect}(\mathcal{F}, A_\Delta, k)$
else
  return $\textsc{PortfolioSelect}(\mathcal{F}, k, \mu)$
end if

in one-to-one correspondence with the unrooted labeled trees on those n terminals. The routine enumerates all $n^{n-2} = 2^{\Theta(n \log n)}$ codes, decodes each tree, orients it at r, remeasures it, and retains the nondominated pairs. This is exactly the terminal-only feasible domain; trees with added Steiner vertices are outside both the enumeration and our problem definition. For example, on terminals $\{0, 1, 2, 3\}$, $(1, 1) \leftrightarrow \{(1, 0), (1, 2), (1, 3)\}$, the star centered at terminal 1. The growth rate explains the small cutoff.

The second exact mechanism is a polynomial dynamic program with quadratic memory. Fix the rooted MST T_M and a radius cap D. Cut MST edges into connected components; retain intracomponent edges, keep the root component centered at r, and attach every other component C by one edge (r, c) to a chosen terminal $c \in C$. Feasibility requires

$$d(r, c) + \max_{v \in C} d_{T_M}(c, v) \leq D. \tag{8}$$

If $F[u, c]$ is the minimum net change inside the rooted subtree T_u while u remains in an open component centered at c, and

$$G[v] = \min_{z \in T_v} \{d(r, z) + F[v, z]\},$$

then a child v containing c must be kept. Every other child contributes

$$\min\{F[v, c],\, G[v] - d(u, v)\}. \tag{9}$$

The $O(n^2)$ states and transitions enumerate exactly the feasible fixed-MST centered family. Reconstruction gives the minimum-length member under cap D, whose radius is rechecked. This is an exact construction within a restricted family, not a stronger universal approximation theorem.

The bounded search operators have concrete topological meanings:

- **Reparent.** Cut the parent edge of a rooted subtree and attach its root to a vertex outside that subtree.
- **Component exchange.** Delete one tree edge and reconnect the two components through any endpoint pair, allowing the detached component to be reoriented.
- **Cap repair.** Under a fixed requirement $R \leq D$, accept only a modification that lowers L.
- **Depth compression.** Using the old root-distance labels, choose shorter valid parent edges while checking that neither L nor R increases.
- **Perturb and descend.** Make a deterministic, coordinate-seeded reparenting perturbation, then resume greedy improvement.

A greedy move is accepted only when reconstruction improves its specified exact score. Total-detour exchange uses $L + \sum_v d_T(r, v)$, not the maximum radius R, as an auxiliary score; the initial perturbation may deliberately worsen its seed. Every resulting tree is nevertheless remeasured and Pareto-filtered in exact (L, R). The most expensive complete component pass is cubic and is confined to the small intensive tier.

5.4 One certified mode, three empirical modes

Every mode constructs and exactly remeasures the balanced height partition A_Δ. The certified selector reserves that tree or an exact Pareto dominator. An empirical selector may trade it away when a small portfolio cap makes frontier coverage more valuable. Consequently, only CERTIFIED carries the universal guarantee in (7); the other three modes continue from the same construction pool but make empirical runtime–quality claims.

- **FAST** uses expanded deterministic anchors, centered candidates and bounded repair, while keeping the search shallow.
- **BALANCED** spends a larger fixed budget on the primary search branch for small instances and reverts to the scalable policy above that tier.
- **QUALITY** adds the full intensive branch, a separately budgeted companion branch on selected moderate-size instances, richer cap schedules, and exact labeled-tree enumeration at the small cutoff.

On selected moderate-size instances, the companion branch reruns the same improvement machinery with centered constructions disabled. This prevents a strong centered seed from consuming the finite seed and beam budget before a structurally different topology is even explored. Finally, the exact frontier is ordered by (L, R, parent). If more than k trees remain, deterministic selection preserves the two endpoints, and admits marked lower-mode witnesses while capacity remains, then fills the largest exact area gaps. Only CertifiedSelect reserves the factor-two witness unconditionally. The result is a portfolio whose modes differ in work, not merely in labels.

6 Experimental Evaluation

The experiments ask three different questions: whether the practical portfolio improves recent methods, how much quality is lost when only the certified construction family is used, and whether the four modes produce a stable runtime–quality ordering. These questions require different statistics. A good tree near the diagonal need not cover an entire cost–radius frontier.

6.1 Data, baselines, and metrics

The frozen comparison contains 28 development instances with 8 to 32 terminals, including synthetic, grid, near-line, and extracted industrial nets. Its published-method reference is the nondominated union of classical constructions, the author PD-II implementation swept over 19 parameter values, and terminal-only trees exported from the 2023 MSPD/MSS implementation over its prescribed 4×4 settings. The internal predecessor developed in our approximation investigation is excluded from this reference and is compared separately in Appendix D. Because these 28 instances were visible during development, we use them for regression and direct baseline comparison, not as evidence of out-of-sample generalization.

A separate 50-instance set, not used to configure the final mode ordering, contains uniform, clustered, grid, comb, radial, and near-line geometries through $n = 96$. Scaling uses 30 instances at $n \in \{64, 128, 256, 512, 1024, 2048\}$, with three retained runs per mode and instance. Four instances through $n = 8$ are small enough for exact labeled-tree enumeration via Prüfer codes.

Every output is independently reconstructed from its parent array and remeasured with integer arithmetic. For portfolios P and Q, write

$$P \succeq Q \quad \Longleftrightarrow \quad \forall q \in Q\ \exists p \in P : L(p) \le L(q) \ \wedge\ R(p) \le R(q).$$

We report *dominance* if $P \succeq Q$ but $Q \not\succeq P$, a *tie* if both hold, *mixed* if neither holds, and a *loss* if only $Q \succeq P$. Hypervolume uses the same recorded reference corner and the same M, Δ normalization for both portfolios on each instance.

To measure only the best balanced representative, define the empirical portfolio common factor

$$c_{\text{port}}(I, P) = \min_{T \in P} \max \left\{ \frac{L(T)}{M_I}, \frac{R(T)}{\Delta_I} \right\}. \tag{10}$$

Unlike hypervolume, c_{port} deliberately ignores the breadth of a portfolio.

6.2 Comparison with the frozen union

Figure 4a separates the theorem-first mode from the empirical portfolio modes. Balanced and quality never lose to the published-method reference: each strictly covers it on 23 instances and ties on the remaining five. Their sums of normalized per-instance hypervolume differences are 0.0863 and 0.0863, respectively. The small difference is consistent with their intended roles: quality spends additional work to recover a few frontier points, rather than changing the objective. Fast is also positive in aggregate hypervolume (0.0654), with 12 dominances, six ties, nine mixed cases, and one loss.

The certified portfolio is covered by the published reference on 26 instances, is mixed on one, and ties on one; its aggregate hypervolume difference is −0.7478. The approximation theorem is therefore not a frontier-dominance result. In direct comparisons, balanced and quality each dominate PD-II on 25 instances and tie it on three; each dominates the terminal MSPD/MSS portfolio on 23 and ties it on five. The union is the stricter comparison because a candidate must cover the complementary points supplied by all published methods at once.

6.3 What does the certificate cost?

The right panel of Figure 4 uses Eq. (10). On the 28 development instances, the certified portfolio has mean $c_{\text{port}} = 1.21$, median 1.24, and range 1.00–1.53. Its nondominated output contains one to six trees, with median three. The much denser reference union has mean 1.06 and maximum 1.18.

The apparent tension with the left panel is informative. A five-threshold height sweep often finds a respectable point near the diagonal, far inside its universal ceiling of two. It does not, however, populate the shoulders between the MST and shortest-path endpoints. Empirical search improves the central compromise and, more decisively, fills those missing regions. The theorem construction is thus a practical core, but not a substitute for the frontier engine.

The same distinction remains on held-out data. There the certified portfolio has mean $c_{\text{port}} = 1.25$, median 1.22, and maximum 1.97. The factor-two witness is independently reconstructed on all 78 development and held-out instances: 66 portfolios contain the original balanced height-partition tree and 12 contain an exactly remeasured Pareto dominator.

6.4 Mode ordering and scaling

On the 50 held-out instances, balanced strictly covers fast on 21 and ties it on 29; quality strictly covers balanced on nine and ties it on 41. Fast strictly covers the certified portfolio on 44, ties it on three, and is incomparable on three; it never loses. Additional empirical budget therefore gives a monotone portfolio hierarchy, while CERTIFIED remains a separate theorem-first endpoint.

All 360 scaling invocations completed. At $n = 2048$:

mode	median ms	max ms	peak MiB
certified	27.1	32.5	2.48
fast	140.8	182.8	3.66
balanced	142.2	220.4	3.75
quality	182.1	302.3	4.05

The source is built with portable `-O3 -DNDEBUG` settings and uses one thread; no architecture-specific compiler flag is enabled.

On the held-out set, the certified mode takes a median 6.9 ms. Fast, balanced, and quality take 13.0, 62.2, and 63.3 ms. These are absolute candidate times. The published artifacts for the recent baselines do not contain same-machine timings under the common terminal-only objective, so we do not form a runtime ratio from their paper-reported numbers.

6.5 Exactness and component evidence

Quality reproduces the complete exact frontier on all four small instances for which every labeled tree was enumerated. The release also passes 26 unit and property tests, independent certificate reconstruction, and exact remeasurement of every reported tree.

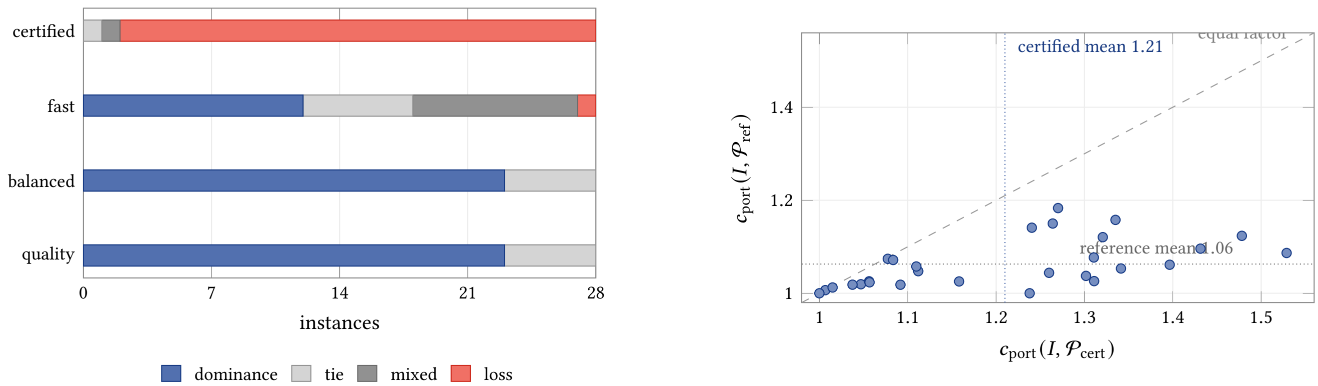

Figure 4: Two views of the 28-instance comparison with the published-method union. The stronger practical modes cover the full union, whereas the theorem-first portfolio does not (left). Nevertheless, its best balanced tree lies well inside the factor-two guarantee; each point at right compares its empirical common factor with that of the denser published portfolio. The panels measure frontier breadth and one-point balance, respectively.

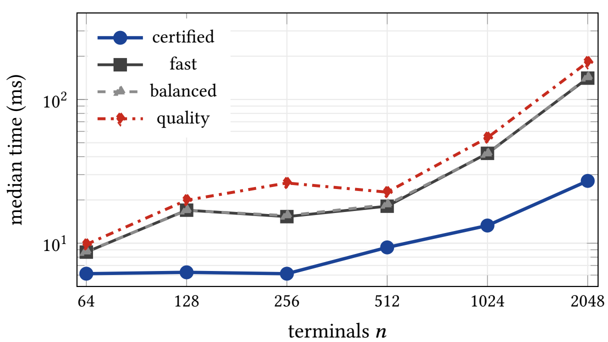

Figure 5: Single-thread end-to-end median time on the scaling set. Each point aggregates three runs for each of five geometries; process startup and serialization are included.

Table 1: Compact quality-mode ablation on the 28-instance frozen union.

Variant	Dom.	Tie	Mix.	Loss	$\sum \Delta$HV
construction only	1	9	8	10	−0.0589
scalable tier only	12	7	9	0	+0.0665
no deep search	20	6	2	0	+0.0840
no component exchange	23	5	0	0	+0.0857
reduced starts	23	5	0	0	+0.0862
HP-RCRST quality	23	5	0	0	**+0.0863**

Table 1 confirms that constructions alone do not explain the full result: they fail to cover the published union on 18 instances. The scalable-only policy makes the largest single jump, and deep search removes the remaining losses. Component exchange and additional starts then recover smaller complementary regions. The ablations are not additive, because constructions are also search seeds. We return to that interaction in Section 7.

7 Discussion

7.1 What is settled, and what is not

The reduction settles the original terminal-only Manhattan decision problem only at the level of ordinary NP-completeness. Because its coordinates scale with the numerical Partition instance, it leaves strong hardness and pseudopolynomial exact algorithms open. The approximation theorem is similarly specific. It gives one tree with $\boldsymbol{\rho} \preceq (2, 2)$ in every metric instance, and the arbitrary-metric construction shows that this line of reasoning cannot force a smaller universal common factor for one tree. It does not rule out a below-two theorem using planar L_1 structure, nor an approximate-Pareto guarantee in which different portfolio members cover different comparison trees.

These boundaries matter because exact optimization, geometric approximation, and portfolio design are different questions. Earlier routing work often met all three through the same construction; the present results separate them.

7.2 Why two exact routes converged

The two polynomial-time investigations received sharply different information. One saw only the problem, benchmark, and evaluator; the other saw the accumulated references and base code. Nevertheless, both reached a parent-arc master with exact rational arithmetic, lazy rooted-connectivity cuts, and path or potential cuts for radius violations.

The convergence was structural rather than social. Once a terminal tree is rooted, its combinatorial content is exactly one parent choice for each nonroot terminal, so total length is linear in the selected parent arcs. Radius has an equally canonical local form: a selected arc adds a nonnegative detour increment to the child's root-path potential. The two remaining global failures are root disconnection and excessive detour on a parent chain. The first exposes a minimum-cut certificate; the second exposes a forbidden path, projected potential, or flow inequality. The shared architecture is therefore the decomposition induced by the problem.

Example 7.1 (Independent witnesses to the same obstruction). The blind condition found a five-terminal frontier containing the neighboring points

$$(L, R) \in \{(27, 12), \boxed{(26, 13)}, (22, 15)\}.$$

For the boxed point to minimize $L + \lambda R$, comparison with the other two points requires both $\lambda \leq 1$ and $\lambda \geq 2$. Independently, the full-context condition found

$$(L, R) \in \{(51, 20), \boxed{(42, 23)}, (34, 24)\},$$

whose boxed point requires both $\lambda \leq 3$ and $\lambda \geq 8$. The instances and incompatible intervals are different. What recurs is the same geometric obstruction: a rectangularly feasible Pareto point can lie above the lower convex envelope visible to every weighted sum.

The common stopping point has the same explanation. Rooted-subset, overlong-path, conflict, and tightened potential inequalities all left small fractional gaps. General integer cuts can force finite exact progress, but polynomial separation in one round does not bound cutting-plane rank, the total number of pivots, or the bit width accumulated across rounds. The two conditions did not independently prove polynomiality; they independently located the same global integrality defect.

Their negative results were still algorithmically selective. Scalarization survived because it cheaply locates supported frontier trees, although unsupported points rule it out as a complete decision method. Local exchange survived because each accepted move improves a checked incumbent, although explicit dominated local optima rule it out as a proof of completeness. The cutting-plane engines themselves were not imported into HP-RCRST. They supplied formulations, audit targets, and warnings about uncontrolled exact-search tails.

7.3 Why the counterexample changed the accounting unit

The approximation investigation first sought an individual-terminal $(2, 2)$ invariant for threshold and LAST-like traversals. It survived more than half a million exhaustive, random, and structured tests before failing on a 58-terminal subdivided broom. Subdivision preserves the long route but splits its removable weight among many small terminal edges. A root shortcut then replaces only one small edge, so charges that looked local on coarse instances accumulate without a global budget.

The height partition changes the unit of accounting. A shortcut is charged to the entire residual component that caused the height threshold to fail; different shortcuts receive disjoint residuals. The proof controls the maximum global radius actually used by the objective, rather than promising the stronger per-terminal stretch statement that the broom falsifies. The counterexample in Example 3.1 did not merely change a constant. It identified the wrong object being amortized.

7.4 A good central tree is not a good frontier

The certified experiment gives a more nuanced answer than either "the theorem won" or "the theorem was only decorative." On the 28 development instances, the small certified portfolio has

$$\text{mean } c_{\text{port}} = 1.21, \qquad \max c_{\text{port}} = 1.53,$$

where c_{port} is the best common factor in the portfolio. On the 50 held-out instances the corresponding values are 1.25 and 1.97. Thus the observed balanced point is usually well inside the worst-case ceiling of two.

The published-method reference nevertheless covers the certified portfolio on 26 of 28 instances; one case is mixed and one is a tie. This is not a contradiction. The common factor asks for one point near the diagonal; hypervolume and portfolio coverage reward a sequence of useful points from the MST end to the shortest-path end. A rooted MST plus a handful of height thresholds is naturally good at the first task. Each threshold resets exactly when accumulated residual height crosses H, so the sweep samples a one-dimensional structural envelope already aligned with length and radius. The same one-dimensional family has too few topological degrees of freedom to fill both shoulders of an empirical frontier.

Example 7.2 (The same MST length, a different rooted geometry). On `grid_5x5`, $M = 416$ and $\Delta = 72$. The certified fixed-MST sweep returns

$$(416, 140), \qquad (515, 89), \qquad (763, 72),$$

whose best common factor is 1.24. The published reference and all three empirical modes instead contain $(416, 72) = (M, \Delta)$, which dominates the entire certified portfolio. Several MSTs have length 416, but their rooted radii differ. Cutting and reattaching one canonical MST explores thresholds on that topology; it does not explore alternative MST tie structures. Here a topology change, not a finer height sweep, closes the frontier in this extreme tie case.

This also clarifies what search contributes. It does not rescue a collapsed certificate: the certified balanced point is already useful and inexpensive. But it is more than a cosmetic refinement. Constructions alone fail to cover the published reference on 18 of 28 instances, whereas the final quality mode has no loss or mixed case and 23 strict dominances. Search improves the central compromise and, more decisively, articulates the rest of the selectable tradeoff curve. The height construction is therefore the algorithmic spine inherited by the final solver. The certified mode reserves its witness; the empirical modes begin from the same construction pool and add search, but may trade that witness away during portfolio truncation.

The certificate is also inexpensive inside the released solver. Once the dense MST is available, the certified mode performs only five linear postorder partitions and exact checks. At $n = 2048$, its median end-to-end time is 27 ms, compared with 141–182 ms for the three empirical modes. This is an internal mode comparison, not a cross-paper runtime claim; the published artifacts do not provide timings under the same objective and measurement boundary.

7.5 Why the final solver kept some ideas and not others

The final condition could inspect every earlier result, but full context did not produce a union of all earlier algorithms. A component survived when it could be assigned a bounded and verifiable role:

- the hardness reduction fixed the claim boundary but supplied no runtime mechanism;
- height partitioning was cheap enough to become a separately checked certified mode;
- star, MST, PD, BRBC, and KRY/LAST constructions became endpoints and diverse search seeds, as well as regression witnesses;
- exactness was retained only where its work was explicitly bounded: complete labeled-tree enumeration for $n \leq 8$ and the fixed-MST centered dynamic program through its deployment cutoff;
- component exchange, reparenting, cap repair, and compression were retained only after empirically checked improvements;
- the earlier rational cutting-plane engines were not ported, because their unresolved tails conflicted with the production role.

The most important transfer from the exact investigations was therefore negative knowledge: which scalarizations miss feasible rectangles, which local minima support no completeness claim, and where rational exact search loses a polynomial bound. Broader information narrowed claims and deployment domains at least as much as it supplied constructions.

Example 7.3 (Search changed the trees, not only their number). On `uniform_n30_s2`, construction alone produces ten nondominated trees; quality mode produces 33. The difference is not merely denser sampling of one curve. At radius 1967, checked search reduces length from 5358 to 5179; at radius 1351, it reduces length from 6370 to 5510. Both portfolios begin at the same $(5115, 3425)$ MST endpoint. The construction pool therefore supplies a reachable region and useful anchors, while exchange search changes the topologies available inside that region.

7.6 Why the components form an ecology

The ablations are non-additive because candidate families participate in a search process, not merely in a final union. With a finite seed and deep-search budget, removing one family also changes the neighborhoods later operators are allowed to explore.

Example 7.4 (A strong family can hide complementary trees). For compactness, abbreviate the three instances by

$$\begin{aligned} U_0 &:= \texttt{uniform\_n30\_s0}, \\ U_2 &:= \texttt{uniform\_n30\_s2}, \\ C_1 &:= \texttt{cluster2\_n30\_s1}. \end{aligned}$$

After exact remeasurement, the companion search without centered candidates retained eight points absent from the primary branch:

$$\begin{aligned} U_0 &: (4330, 1880), (4412, 1757), (4453, 1731), \\ U_2 &: (5127, 3329), \\ C_1 &: (2579, 1676), (2582, 1674), \\ &\quad\ (2594, 1666), (2595, 1665). \end{aligned}$$

Omitting both centered and critical-height constructions still lost one prior witness. The final policy therefore gives the two searches separate budgets and merges them only after both have produced checked frontiers. Mixing all seeds before search would recreate the crowding that exposed these eight points.

For the same reason, provenance tags are terminal labels rather than causal weights. Of 263 public-set output trees, 203 are tagged as exchange search, whereas only 17 retain a PD, LAST, or BRBC tag and seven retain a structural construction tag. An anchor may disappear after several exchanges while remaining necessary to reach the final topology. Conversely, deleting a frequently returned family can free enough budget to improve another branch. The pipeline is better understood through interactions among seeds, operators, and finite budgets than through a ranking of standalone heuristics.

7.7 The research object was the transition

The most revealing observations in this study are not well described by a division between "human" and "machine" work. The blind and full-context conditions converged because the formulation forced them to; the approximation condition advanced when a long-lived conjecture was falsified; and the final solver used earlier failures more directly than it used earlier code. In each case, progress occurred at a transition between inspectable objects:

$$\begin{aligned} \text{conjecture} \longrightarrow \text{inequality} \longrightarrow \text{counterexample} \\ \longrightarrow \text{revised theorem} \longrightarrow \text{checked construction.} \end{aligned}$$

That chain is more informative than the number of generated proposals.

It also suggests a concrete research regime for older algorithmic problems. The favorable cases are not simply those with a short statement. They have a white-box objective, inexpensive exact falsification on small instances, several plausible formulations, limited hidden experimental infrastructure, and a literature precise enough to define what remains open. Information boundaries can then be varied as an experimental condition: a blind attempt tests rediscovery and structural convergence; a full-context attempt tests whether inherited knowledge actually changes the bottleneck; an adversarial condition tests the claims produced by both.

Agents for formal mathematics, evaluator-guided program discovery, and self-evolving EDA systems emphasize different parts of this regime [11, 23, 27, 30]. Prim–Dijkstra required all three forms of evidence: a proof that could survive reconstruction, a counterexample that could change the proof, and an implementation whose frontier could survive a frozen evaluator. The public artifact exposes a reproducible subset of those objects, with the boundary stated in Section 3; it is not offered as a complete hidden trace.

The older literature was usable because its definitions, constructions, and open question could be reconstructed without changing the model. In this study that precision let old claims become evaluator cases, proof obligations, and counterexample searches. It is a practical precondition for the research regime above: without stable definitions and checkable objects, additional agent effort would only multiply incompatible interpretations.

8 Conclusion

This revisit settles the terminal-only Manhattan decision problem, gives its global cost–radius tradeoff a direct $(2, 2)$ construction, and carries that construction into HP-RCRST. The process record adds three concrete observations: two independently constrained exact attempts reached the same integrality barrier; one 58-terminal counterexample redirected the approximation proof; and the final solver retained the theorem-bearing construction without importing unresolved exact-search tails. Together they form an audited case in which language-model agents helped move an older algorithmic question from reconstruction to proof and implementation, while proofs, counterexamples, and a frozen evaluator remained the standards of evidence.

References

[1] Charles J. Alpert, Wing-Kai Chow, Kwangsoo Han, Andrew B. Kahng, Zhuo Li, Derong Liu, and Sriram Venkatesh. 2018. Prim-Dijkstra Revisited: Achieving Superior Timing-driven Routing Trees. In *Proceedings of the 2018 International Symposium on Physical Design*. ACM, 10–17. https://doi.org/10.1145/3177540.3178239

[2] Charles J. Alpert, T. C. Hu, Dennis J.-H. Huang, Andrew B. Kahng, and David R. Karger. 1995. Prim-Dijkstra Tradeoffs for Improved Performance-Driven Routing Tree Design. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 14, 7 (1995), 890–896. https://doi.org/10.1109/43.391737

[3] Iago A. Carvalho and Amadeu A. Coco. 2023. On Solving Bi-Objective Constrained Minimum Spanning Tree Problems. *Journal of Global Optimization* 87, 1 (2023), 301–323. https://doi.org/10.1007/s10898-023-01295-8

[4] Gengjie Chen and Evangeline F. Y. Young. 2020. SALT: Provably Good Routing Topology by a Novel Steiner Shallow-Light Tree Algorithm. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 39, 6 (2020), 1217–1230. https://doi.org/10.1109/TCAD.2019.2894653

[5] Otfried Cheong and Changryeol Lee. 2013. Single-Source Dilation-Bounded Minimum Spanning Trees. *International Journal of Computational Geometry & Applications* 23, 3 (2013), 159–170. https://doi.org/10.1142/S0218195913500052

[6] Jingsheng Cong, Andrew B. Kahng, Gabriel Robins, Majid Sarrafzadeh, and C. K. Wong. 1992. Provably Good Performance-Driven Global Routing. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 11, 6 (1992), 739–752. https://doi.org/10.1109/43.137519

[7] Siad Daboul, Stephan Held, Bento Natura, and Daniel Rotter. 2023. Global Interconnect Optimization. *ACM Transactions on Design Automation of Electronic Systems* 28, 5 (2023), 72:1–72:24. https://doi.org/10.1145/3587044

[8] Michael Elkin and Shay Solomon. 2015. Steiner Shallow-Light Trees Are Exponentially Lighter than Spanning Ones. *SIAM J. Comput.* 44, 4 (2015), 996–1025. https://doi.org/10.1137/13094791X

[9] Josefine Foos, Stephan Held, and Yannik Kyle Dustin Spitzley. 2023. Tighter Approximation for the Uniform Cost-Distance Steiner Tree Problem. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (Leibniz International Proceedings in Informatics)*, Vol. 275. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 19:1–19:16. https://doi.org/10.4230/LIPIcs.APPROX/RANDOM.2023.19

[10] Michael R. Garey and David S. Johnson. 1979. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman.

[11] Google DeepMind. 2025. AlphaEvolve: A Gemini-Powered Coding Agent for Designing Advanced Algorithms. https://deepmind.google/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/.

[12] Luis Gouveia, Ana Paias, and Dushyant Sharma. 2008. Modeling and Solving the Rooted Distance-Constrained Minimum Spanning Tree Problem. *Computers & Operations Research* 35, 2 (2008), 600–613. https://doi.org/10.1016/j.cor.2006.03.022

[13] Stephan Held, Dirk Müller, Daniel Rotter, Rudolf Scheifele, Vera Traub, and Jens Vygen. 2018. Global Routing with Timing Constraints. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 37, 2 (2018), 406–419. https://doi.org/10.1109/TCAD.2017.2697964

[14] Stephan Held and Edgar Perner. 2025. Cost-Distance Steiner Trees for Timing-Constrained Global Routing. In *Proceedings of the 62nd ACM/IEEE Design Automation Conference*. IEEE, 1–6. https://doi.org/10.1109/DAC63849.2025.11132669

[15] Jan-Ming Ho, D. T. Lee, Chia-Hsiang Chang, and C. K. Wong. 1991. Minimum Diameter Spanning Trees and Related Problems. *SIAM J. Comput.* 20, 5 (1991), 987–997. https://doi.org/10.1137/0220060

[16] Andrew B. Kahng, Shreyas Thumathy, and Mingyu Woo. 2023. An Effective Cost-Skew Tradeoff Heuristic for VLSI Global Routing. In *2023 24th International Symposium on Quality Electronic Design*. IEEE, 1–8. https://doi.org/10.1109/ISQED57927.2023.10129350

[17] Ardalan Khazraei and Stephan Held. 2021. An Improved Approximation Algorithm for the Uniform Cost-Distance Steiner Tree Problem. In *Approximation and Online Algorithms (Lecture Notes in Computer Science)*, Vol. 12806. Springer, 189–203. https://doi.org/10.1007/978-3-030-80879-2_13

[18] Samir Khuller, Balaji Raghavachari, and Neal E. Young. 1995. Balancing Minimum Spanning Trees and Shortest-Path Trees. *Algorithmica* 14, 4 (1995), 305–321. https://doi.org/10.1007/BF01294129

[19] Hung Le, Shay Solomon, Cuong Than, Csaba D. Tóth, and Tianyi Zhang. 2026. Approximating Euclidean Shallow-Light Trees. In *42nd International Symposium on Computational Geometry (Leibniz International Proceedings in Informatics)*, Vol. 367. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 71:1–71:16. https://doi.org/10.4230/LIPIcs.SoCG.2026.71

[20] Adam Meyerson, Kamesh Munagala, and Serge A. Plotkin. 2008. Cost-Distance: Two Metric Network Design. *SIAM J. Comput.* 38, 4 (2008), 1648–1659. https://doi.org/10.1137/050629665

[21] Jaewon Oh, Iksoo Pyo, and Massoud Pedram. 1997. Constructing Minimal Spanning/Steiner Trees with Bounded Path Length. *Integration, the VLSI Journal* 22, 1–2 (1997), 137–163. https://doi.org/10.1016/S0167-9260(97)00009-6

[22] OpenAI. 2026. A Proof of the Cycle Double Cover Conjecture. https://cdn.openai.com/pdf/04d1d1e4-bc75-476a-97cf-49055cd98d31/cdc_proof.pdf.

[23] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. P. Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. 2024. Mathematical Discoveries from Program Search with Large Language Models. *Nature* 625 (2024), 468–475. https://doi.org/10.1038/s41586-023-06924-6

[24] Hussein F. Salama, Douglas S. Reeves, and Yannis Viniotis. 1997. The Delay-Constrained Minimum Spanning Tree Problem. In *Proceedings of the Second IEEE Symposium on Computers and Communications*. IEEE, 699–703. https://doi.org/10.1109/ISCC.1997.616089

[25] Dae Young Seo, D. T. Lee, and Tien-Ching Lin. 2009. Geometric Minimum Diameter Minimum Cost Spanning Tree Problem. In *Algorithms and Computation (ISAAC 2009) (Lecture Notes in Computer Science)*, Vol. 5878. Springer, 283–292. https://doi.org/10.1007/978-3-642-10631-6_30

[26] Shay Solomon. 2015. Euclidean Steiner Shallow-Light Trees. *Journal of Computational Geometry* 6, 2 (2015), 113–139. https://doi.org/10.20382/jocg.v6i2a7

[27] Terence Tao et al. 2026. From Solvers to Research: Large Language Model-Driven Formal Mathematics at the Research Frontier. arXiv:2607.07779 [cs.AI]

[28] Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. 2024. Solving Olympiad Geometry without Human Demonstrations. *Nature* 625, 7995 (2024), 476–482. https://doi.org/10.1038/s41586-023-06747-5

[29] Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan J. Prenger, and Animashree Anandkumar. 2023. LeanDojo: Theorem Proving with Retrieval-Augmented Language Models. In *Advances in Neural Information Processing Systems*, Vol. 36.

[30] Cunxi Yu and Haoxing Ren. 2026. Autonomous Evolution of EDA Tools: Multi-Agent Self-Evolved ABC. In *Proceedings of the 63rd ACM/IEEE Design Automation Conference*. https://doi.org/10.1145/3770743.3804221 arXiv:2604.15082

A Complete Weak NP-Completeness Proof

This appendix supplies the details suppressed in Theorem 4.1. We reduce from binary PARTITION. Its input is a list of positive integers $a_1, \ldots, a_n$; the question is whether some subset sums to $S/2$, where $S = \sum_i a_i$. This problem is NP-complete under polynomial-time many-one reductions [10].

A.1 Membership in NP

A certificate lists $n-1$ unordered pairs of terminal indices. A verifier rejects invalid indices, loops, and any edge whose endpoints are already joined in a disjoint-set structure; after the last edge it checks that one component remains. It then computes every edge length $|x_i - x_j| + |y_i - y_j|$, sums them, and performs one rooted traversal to compute all root distances.

If N is the complete binary input length, a coordinate difference, an edge, and a sum of at most $n-1$ edges have $O(N + \log n) = O(N)$ bits. There are polynomially many exact additions, comparisons, and graph operations. The empty certificate for $n = 1$ gives $L = R = 0$. Consequently the decision problem belongs to NP.

A.2 The reduction

Given $a_1, \ldots, a_n$, put

$$S = \sum_i a_i, \qquad h = 10S, \qquad t = 4S, \qquad X = 2hn + t.$$

Create

$$s_i = (2hi, 0) \quad (0 \le i \le n), \qquad u_i = (2hi - 3a_i, a_i) \quad (1 \le i \le n),$$

and $z = (X, 0)$, with root s_0. The budgets are

$$B = X + 3S, \qquad D = X + S. \tag{11}$$

Since

$$x(u_i) - x(s_{i-1}) = 2h - 3a_i \ge 17S > 0, \qquad x(s_i) - x(u_i) = 3a_i > 0,$$

the strict order is

$$s_0, u_1, s_1, u_2, s_2, \ldots, u_n, s_n, z. \tag{12}$$

All terminals are distinct. Every coordinate and budget is $O(nS)$, so each has $O(\log n + \log S)$ bits and the mapping has polynomial encoding length.

A.3 Forward implication

Suppose $I \subseteq [n]$ has $A = \sum_{i \in I} a_i = S/2$. Include $u_i s_i$ for every i; include $s_{i-1}u_i$ if $i \in I$ and $s_{i-1}s_i$ otherwise; finally include $s_n z$. The construction joins one new block at a time and is a tree. Its relevant lengths are

$$d(s_{i-1}, u_i) = 2h - 2a_i, \quad d(s_{i-1}, s_i) = 2h,$$
$$d(u_i, s_i) = 4a_i, \qquad d(s_n, z) = t.$$

A selected block contributes $2h + 2a_i$ to L, and an unselected block contributes $2h + 4a_i$. Hence

$$L = X + 2A + 4(S - A) = X + 3S = B. \tag{13}$$

The root–z path uses $u_i s_i$ exactly for $i \in I$, and therefore

$$d_T(s_0, z) = X + 2A = X + S = D. \tag{14}$$

Every vertex on that path is within D. If $i \notin I$, the only off-path vertex in the block is u_i. With $A_i = \sum_{j \in I, j \le i} a_j$,

$$d_T(s_0, u_i) = 2hi + 2A_i + 4a_i \le 2hn + S + 4S = 2hn + t + S = D. \tag{15}$$

Thus the produced tree is feasible.

A.4 Reverse implication

Let T be any feasible tree in the complete Manhattan graph, and let P be its unique s_0-to-z path. Give a tree edge multiplicity two if it lies on P and one otherwise. The weighted sum is

$$F = L(T) + d_T(s_0, z) \le B + D = 2X + 4S. \tag{16}$$

LEMMA A.1 (INTERBLOCK CUT FORCING). *Exactly one tree edge crosses every prefix cut between two consecutive blocks in (12), and that edge lies on P.*

PROOF. For an open vertical gap between consecutive terminals, let c_T and c_P be the numbers of crossing tree and path edges. Connectivity gives $c_T \ge 1$, while the path from the leftmost to the rightmost terminal gives $c_P \ge 1$. An abstract complete-metric edge contributes its horizontal length once to every gap between its endpoints, so the horizontal part of F is exactly the sum of each gap width times $c_T + c_P$. The baseline over all gaps is $2X$.

The gap preceding block i has width at least $17S > 4S$. A third crossing would make the horizontal contribution exceed (16). The final gap has width $4S$; a third crossing there would make its horizontal contribution at least $2X + 4S$, while the vertical contribution is strictly positive because some u_i has positive height. Thus $c_T + c_P = 2$ on every interblock cut, so $c_T = c_P = 1$. □

LEMMA A.2 (FORCED TOPOLOGY). *The tree has exactly one edge between each pair of consecutive blocks, no other interblock edge, and every internal edge $u_i s_i$. All interblock edges lie on P.*

PROOF. The unique edges of two different nested prefix cuts must be distinct: deleting one tree edge produces one fixed root-side component, not two different prefixes. An edge that skips a block would cross two such cuts and would have to be their common unique edge. Therefore the $n+1$ interblock edges are distinct and join consecutive blocks. A tree on $2n + 2$ vertices has $2n + 1$ edges, leaving exactly n intrablock edges. Each two-vertex block has only the edge $u_i s_i$, so all of them are forced. Lemma A.1 places every interblock edge on P. □

Let W be the total length of the interblock edges, let

$$I = \{i : u_i s_i \text{ lies on } P\}, \qquad A = \sum_{i \in I} a_i.$$

Lemma A.2 gives the exact identities

$$L(T) = W + 4S, \qquad d_T(s_0, z) = W + 4A. \tag{17}$$

LEMMA A.3 (BRIDGE INEQUALITY). *$W + 2A \ge X$.*

PROOF. Orient P from s_0 to z. Split I into I_+, whose internal edge is traversed $u_i \to s_i$, and I_-, traversed in reverse; write A_+ and A_- for their item sums. If H_b, V_b are the total horizontal and absolute vertical changes on the bridges, then $W = H_b + V_b$. Telescoping the signed x-change along P yields

$$H_b = X - 3A_+ + 3A_-.$$

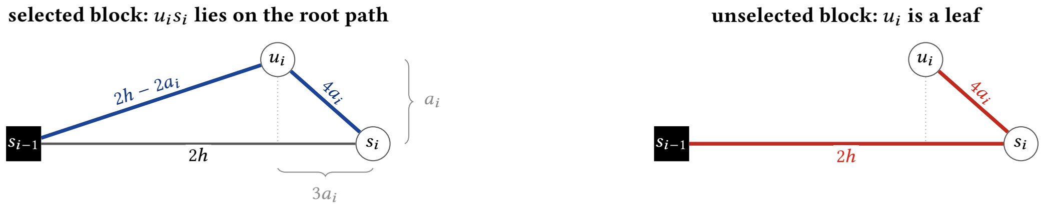

Figure 6: The two canonical block choices in the forward reduction. Blue and muted-red paths are combinatorial terminal edges weighted by Manhattan distance; the dotted projection is only a coordinate guide. The reverse proof does not assume either drawing and starts from an arbitrary tree in the complete terminal metric.

Telescoping signed y-change yields bridge signed change $A_+ - A_-$, so $V_b \geq A_+ - A_-$. Therefore

$$\begin{aligned} W + 2A &= H_b + V_b + 2(A_+ + A_-) \\ &\geq X + 4A_- \geq X. \end{aligned}$$

Reverse traversal only adds slack. □

Using (17), Lemma A.3, and the cost budget,

$$X - 2A \leq W \leq B - 4S = X - S,$$

so $A \geq S/2$. The path-distance budget gives

$$X + 2A \leq W + 4A = d_T(s_0, z) \leq D = X + S,$$

so $A \leq S/2$. Thus $A = S/2$, and I is a valid Partition witness. This argument quantifies over every complete Manhattan edge, including edges with noncanonical consecutive-block endpoints.

A.5 Classification

The construction proves NP-hardness and, with membership in NP, NP-completeness. Its coordinates are proportional to S, so the reduction is numerical and establishes weak NP-hardness. It neither proves strong NP-hardness nor rules out a pseudopolynomial algorithm.

B Height Partition and the Factor-Two Boundary

B.1 Lower bounds and edge-cut facts

Let d be any finite metric, let r be the root, and define

$$M = \mathrm{MST}(V, d), \qquad \Delta = \max_{v \in V} d(r, v).$$

Every terminal tree T satisfies $M \leq L(T)$ by MST optimality and $\Delta \leq R(T)$ by the triangle inequality along each root path. These are the two independent lower bounds used by the approximation ratio.

Fix an MST T_M rooted at r. For every parent edge (u, v) of weight e_v, deleting the edge exposes a cut containing v but not r. The complete-metric edge (r, v) crosses this cut, so the MST cut property gives

$$e_v \leq d(r, v) \leq \Delta. \tag{18}$$

The nonnegative net cost of replacing (u, v) by (r, v) is therefore

$$a_v = d(r, v) - e_v \leq \Delta. \tag{19}$$

B.2 Residual invariant

The mathematical construction permits any real $H > 0$; the integer implementation evaluates exact integer thresholds. Process vertices in postorder. At the time a vertex v is processed, its *open residual* C_v consists of v and precisely those descendant residuals whose joining edges have been retained. Let

$$W_v = \sum_{e \in E(C_v)} d(e), \qquad h_v = \max_{x \in C_v} d_{C_v}(v, x).$$

The algorithm retains (u, v) iff

$$e_v + h_v \leq H. \tag{20}$$

If retained, C_v is merged into C_u; otherwise it is closed, (u, v) is removed, and (r, v) is installed.

Lemma B.1 (Residual height). *Every open or closed residual has height at most H. The final graph is a terminal spanning tree.*

Proof. A leaf residual has height zero. Inductively, every retained child residual has height from its parent at most $e_v + h_v \leq H$, so their union with the parent also has height at most H. Cutting an edge closes exactly one connected residual. Starting from a tree, each cut increases the number of components by one and each root shortcut reconnects exactly the newly closed component; no shortcut connects two vertices already in the same residual. The final graph is therefore connected with $n-1$ edges and is a tree. □

Lemma B.2 (Radius). *The returned tree A_H satisfies $R(A_H) \leq \Delta + H$.*

Proof. The root residual is reached without a shortcut and has height at most H. Every other residual is reached through its unique root edge (r, v), of length at most Δ, followed by a residual path of length at most H. Taking the maximum gives the claim. □

B.3 Disjoint charge

For a cut child v, use as its resource the deleted edge together with the edges of its closed residual:

$$Q_v = \{(u, v)\} \cup E(C_v), \qquad w(Q_v) = e_v + W_v.$$

The failed test and $W_v \geq h_v$ imply

$$w(Q_v) = e_v + W_v \geq e_v + h_v > H. \tag{21}$$

Combining (19) and (21),

$$a_v H \leq \Delta\, w(Q_v). \tag{22}$$

Lemma B.3 (Disjointness). *The sets Q_v over all cut children are pairwise edge-disjoint, and their total weight is at most M.*

Proof. When C_v is closed, its edges no longer belong to the open residual passed to any ancestor. A later cut can charge its own joining edge and the edges that remain in its open residual, but none already closed below it. Two cuts in different branches have disjoint subtree edges. Hence no MST edge is charged twice, and all charged edges belong to T_M. □

The output differs from T_M only at the cut children. Summing (22) and using Lemma B.3 yields

$$\begin{aligned}\big(L(A_H) - M\big)H &= H \sum_{v \text{ cut}} a_v \\ &\le \Delta \sum_{v \text{ cut}} w(Q_v) \le M\Delta.\end{aligned} \tag{23}$$

Together with Lemma B.2, this proves

$$L(A_H) \le M\left(1 + \frac{\Delta}{H}\right), \qquad R(A_H) \le \Delta + H. \tag{24}$$

At $H = \Delta$, (24) gives $L \le 2M$ and $R \le 2\Delta$. If $n = 1$, then $M = \Delta = L = R = 0$ and the empty tree is handled separately. For $n > 1$ with distinct terminals, $\Delta > 0$.

The output topology changes only when H crosses one of finitely many residual heights. Nevertheless, (24) holds for every real $H > 0$, giving the continuous certificate envelope plotted in Figure 3. The released mode samples five integer thresholds and always includes the exact $H = \Delta$ member.

B.4 Why metric and MST arguments alone stop at two

Theorem B.4 (Arbitrary-metric one-tree lower bound). *For every $c < 2$, there is a finite positive-integer metric in which every terminal tree A satisfies*

$$\max\{L(A)/M,\ R(A)/\Delta\} > c.$$

Proof. The case $c < 1$ is immediate, so write $\varepsilon = 2 - c \in (0, 1]$. Choose

$$q = \max\{2, \lceil 4/\varepsilon \rceil - 1\}, \qquad m = q + 1.$$

Form a generating graph with a unit-edge trunk of length $2q + 1$ from root r to a branch vertex, and attach m disjoint unit-edge arms of length $q+1$. Add an edge of weight q from r to every other vertex, and take the shortest-path metric on all vertices.

The unit trunk-and-arms tree has

$$M = 2q + 1 + m(q + 1), \tag{25}$$

because every nonzero metric distance is at least one and it already has $n - 1$ unit edges. The root shortcuts give $\Delta \le q$, while remote vertices have distance exactly q, hence $\Delta = q$.

Take an arbitrary complete-metric terminal tree A. Expand each edge into a shortest path in the generating graph, take the union, and then take a root shortest-path tree B of that union. This operation cannot increase either objective:

$$L(B) \le L(A), \qquad R(B) \le R(A). \tag{26}$$

If $R(A) > 2q$, then $R(A)/\Delta > 2 > c$. Otherwise every path in B to an arm endpoint has length at most $2q$. A path beginning with the unit trunk has length $3q + 2$; one beginning with a q-edge to the trunk, branch, or another arm has length at least $2q+1$. Thus reaching each arm endpoint requires a distinct q-edge whose nonroot endpoint lies strictly inside that arm. Tree B contains at least m such edges. Its other $n - 1 - m$ edges have weight at least one, so

$$L(A) \ge L(B) \ge M + m(q - 1). \tag{27}$$

Combining the two cases,

$$\max\{L(A)/M, R(A)/\Delta\} \ge 1 + \frac{m(q - 1)}{2q + 1 + m(q + 1)}. \tag{28}$$

The choice of q, m makes the right-hand side exceed $2 - \varepsilon = c$. Indeed $\varepsilon(q + 1) \ge 4$, and direct rearrangement reduces the desired inequality to

$$m > \frac{(1 - \varepsilon)(2q + 1)}{\varepsilon(q + 1) - 2},$$

whose right-hand side is less than $(2q + 1)/2 < q + 1 = m$. The construction is finite; it has

$$n = q^2 + 4q + 3 < 35/\varepsilon^2$$

vertices. □

This theorem is intentionally qualified. The metric need not embed in the Manhattan plane, and the result concerns one tree simultaneously compared with the MST and star lower-bound witnesses. Different members of a portfolio may cover different comparison trees. Neither a planar-L_1 factor below two nor an arbitrary-metric portfolio lower bound follows.

C Algorithmic Contract and Exact Subroutines

This appendix fixes the boundary between the mathematical construction, the bounded empirical search, and the object that is finally emitted. The description follows the released implementation rather than an idealized variant of it.

C.1 Parent arrays and exact remeasurement

Let the terminals be indexed by $V = \{0, \ldots, n - 1\}$, with root $r = 0$. A candidate is a complete parent array

$$p[0] = -1, \qquad p[v] \in V \setminus \{v\} \quad (v \ne 0).$$

The validator first builds the child lists induced by p. A traversal from 0 must visit every terminal exactly once; a repeated visit is a cycle, and an unvisited terminal is disconnected. Hence a successful array contains exactly $n - 1$ arcs and represents one rooted terminal tree.

For every successful array, the validator discards all incremental objective estimates and recomputes

$$\ell_v = |x_v - x_{p[v]}| + |y_v - y_{p[v]}|, \qquad L(p) = \sum_{v \ne 0} \ell_v,$$

and, during the same rooted traversal,

$$q_0 = 0, \qquad q_v = q_{p[v]} + \ell_v, \qquad R(p) = \max_{v \in V} q_v. \tag{29}$$

Exact parent-array duplicates are removed first. After remeasurement, candidates are ordered lexicographically by (L, R, p); equal metric pairs are coalesced, and a monotone scan in increasing L

retains a point precisely when its R is strictly smaller than every preceding radius. Thus neither a construction tag nor an incremental move score can affect feasibility or Pareto dominance.

Arithmetic domain. The certified domain consists of distinct terminals whose two coordinate tokens lie in $[-2^{63}, 2^{63}-1]$. Coordinate differences are widened before subtraction. A Manhattan edge is therefore smaller than 2^{65}. Because the implementation uses an $n < 2^{31}$ terminal index, every $(n-1)$-edge tree length and every root path is smaller than 2^{96}; signed 128-bit accumulators contain (29) exactly. Cross-products used for ratio comparisons, area selection, and certificate inequalities use arbitrary-precision integers. No floating-point value participates in validation, dominance, or witness selection.

The CERTIFIED mode rejects a coordinate token outside the stated domain. For such a token the empirical interfaces return only the valid terminal star fallback, before entering fixed-width metric arithmetic; that fallback is not covered by the certified-mode claim. Duplicate terminals are rejected in every mode. The requested portfolio cap is clamped to $1 \le k \le 64$.

C.2 Certificate-preserving selection

Let C be the exactly remeasured height-partition tree for $H = \Delta$, and let $\mathcal{C}$ contain the MST, the star, and all five height-partition candidates used by the certified mode. By Theorem 4.2,

$$L(C) \le 2M, \qquad R(C) \le 2\Delta. \tag{30}$$

The certified selector is separate from the empirical portfolio selector. If $k = 1$, it returns the parent array of C itself. If $k > 1$, it first forms the exact nondominated frontier $\mathcal{F}$ of $\mathcal{C}$ and chooses

$$C^\star \in \arg \min_{\substack{T\in\mathcal{F}\\ L(T)\le L(C),\ R(T)\le R(C)}} \big(L(C)-L(T)\big) + \big(R(C)-R(T)\big), \tag{31}$$

breaking a remaining tie by the parent array. Selection begins with $C^\star$, adds the two frontier endpoints if space remains, and fills remaining slots by the exact area rule described below. It never deletes $C^\star$.

PROPOSITION C.1 (WITNESS PRESERVATION). *For every supported instance and every $k \ge 1$, the certified output $\mathcal{P}_{\text{cert}}$ contains a tree T satisfying (30). For $k = 1$, that tree has exactly the parent array produced by* HEIGHTPARTITION(T_M, r, Δ).

PROOF. The statement for $k = 1$ is the selector's explicit first branch. For $k > 1$, the finite candidate set contains C, so its nondominated frontier contains either C or a tree that Pareto-dominates C. Consequently the feasible set in (31) is nonempty. The chosen $C^\star$ inherits both inequalities in (30) and is inserted before any downsampling; the certified selector reserves its index throughout truncation. □

There is a second, independent check immediately before output. It rebuilds the canonical MST and C, recomputes $M, \Delta, L(C), R(C)$, remeasures every returned parent array, and verifies that at least one returned pair dominates $(L(C), R(C))$. When $k = 1$, it additionally checks parent-array identity with C. This check does not prove Theorem 4.2; it tests that the executable path has preserved the theorem's witness.

C.3 Exact centered partitioning of a fixed MST

The centered dynamic program is exact for a restricted family that is useful as an empirical anchor. Fix a rooted MST T_M, its edge weights w, and a radius cap D. Delete any subset $Q \subseteq E(T_M)$. The remaining components must retain their MST edges. The component containing r is centered at r; every other component C chooses one center $c_C \in C$ and receives the direct edge (r, c_C). Such a partition has

$$L = M - \sum_{e\in Q} w(e) + \sum_{C\ne C_r} d(r, c_C), \tag{32}$$

and is feasible exactly when

$$d(r, c_C) + d_{T_M}(c_C, v) \le D \quad \text{for every } v \in C, \tag{33}$$

with $c_{C_r} = r$. Here d_{T_M} denotes distance in the fixed MST, not Manhattan distance along a newly optimized tree.

For a vertex u, write T_u for its rooted subtree and $\mathrm{ch}(u)$ for its children. The open state $F_u(c)$ is the minimum net change relative to the retained MST edges inside T_u, conditioned on u belonging to an as-yet-open component centered at c. The center may lie outside T_u, because that component can continue through the parent edge. Set $F_u(c) = +\infty$ whenever

$$d(r, c) + d_{T_M}(c, u) > D. \tag{34}$$

For a feasible state, define the cost of closing a component at u by

$$G_u = \min_{z\in T_u} \{d(r, z) + F_u(z)\}. \tag{35}$$

The postorder recurrence is

$$F_u(c) = \sum_{v\in\mathrm{ch}(u)} \begin{cases} F_v(c), & c \in T_v, \\ \min\{F_v(c),\ G_v - w(u, v)\}, & c \notin T_v, \end{cases} \tag{36}$$

subject to (34). A leaf has value 0 for every state satisfying (34). The optimum length in the centered family is

$$L_{\text{cent}}(D) = M + F_r(r). \tag{37}$$

The first alternative in (36) retains (u, v). The second alternative either retains it, or cuts it, closes the child component using (35), and subtracts the removed MST edge. The direct root edge of a closed component is already included by the term $d(r, z)$ in (35).

THEOREM C.2 (EXACTNESS OF THE CENTERED DP). *For fixed T_M and D, recurrence (34)–(37) returns a minimum-length tree among all feasible centered partitions of T_M. Its reconstruction has radius at most D. With all pairwise tree distances generated in quadratic time, the dynamic program uses $O(n^2)$ time and $O(n^2)$ memory.*

PROOF. Proceed by induction over the rooted subtrees. Fix a feasible state (u, c). If $c \in T_v$, the unique u-to-c path in T_M contains (u, v), so that edge must remain and the child has state (v, c). If $c \notin T_v$, either the child remains in the open component, again giving $F_v(c)$, or (u, v) is cut. In the latter case the entire child subtree is independent of the other children and its top component must choose some center $z \in T_v$; (35) gives its optimum direct attachment and the term $-w(u, v)$ accounts for the cut. The child choices are edge-disjoint, so their net changes add, proving (36).

Conversely, every choice in (36) either keeps or cuts exactly one child edge. Induction reconstructs a valid centered partition, so the recurrence neither omits nor introduces a member of the family. Condition (34), applied at every vertex of every open component, is precisely (33). At the root, (r, r) leaves the root component open without adding a root self-edge, which proves (37) and the radius statement.

Euler intervals decide $c \in T_v$ in $O(1)$ time. There are n^2 open states. Summed over all centers, the child transitions cost $n \sum_u |\operatorname{ch}(u)| = O(n^2)$; computing (35), tree distances, and reconstruction has the same bound. The table of $F_u(c)$ dominates memory. ◻

The restriction in Theorem C.2 is substantive: component interiors must be subtrees of one fixed MST, and each nonroot component uses one direct root attachment. The theorem is neither an exact algorithm for unrestricted RCRST nor an improvement of the universal factor in Theorem 4.2.

C.4 Frozen production policies

All schedules are deterministic functions of n, k, and measured geometry; neither an instance name nor a stored baseline objective enters dispatch. The four public policies are the following.

CERTIFIED. One canonical dense-Prim MST, the star, and height partitions at

$$H \in \{\lfloor \Delta/4 \rfloor, \lfloor \Delta/2 \rfloor, \Delta, 2\Delta, 4\Delta\},$$

with every positive threshold rounded up to at least 1. No perturbation, local search, beam, centered DP, or exponential enumeration is invoked.

FAST. The scalable schedule uses up to three Prim and three Kruskal tie variants, six PD/BRBC/KRY settings, six LAST settings, seven radial settings, four centered-MST caps through $n = 128$, cap repair through $n = 128$, and bounded repair through $n = 40$. Above the applicable cutoffs, the corresponding families are skipped rather than timed out.

BALANCED. For $n \le 32$, this policy runs the centered primary quality branch with 96 deterministic perturbation trials but disables the companion branch. For $n > 32$, it uses the same scalable schedule as FAST. Exact labeled-tree enumeration via Prüfer codes is therefore present in BALANCED for $n \le 8$, because it belongs to the small primary quality engine.

QUALITY. The primary intensive engine uses 96 trials through $n = 32$, with nine LAST and twelve radial settings, critical MST heights, deeper reparent and component neighborhoods, and bounded cap beams. It enumerates the exact labeled-tree frontier for $n \le 8$, applies the centered DP through $n = 128$, and retains the richer cap schedule through $n = 256$. For $26 \le n \le 30$, it unions a separately budgeted companion branch in which the centered construction family is disabled.

Budget terminology. One *perturbation trial* selects a frontier seed, applies one to five coordinate-seeded reparenting moves, and then performs improving component exchanges. A *neighborhood* is the set of trees reachable by one stated move. A $w \times d$ *cap beam* keeps at most w feasible trees, ordered by (L, R, parent), at each of d successive component-exchange rounds.

The companion branch uses only the ratio between the maximum and median edge of the canonical MST. A ratio at least 8 receives 96 clustered trials; a ratio at least 2 receives 32 trials with the deeper companion neighborhood; all other cases receive 32 trials with the shallower neighborhood. The primary and companion defaults are, respectively, 48 and at most 48 reparent seeds, two and at most two *additional* deep reparent layers after the first two neighborhood expansions, 32 and at most 32 component seeds, two and at most two component rounds, and an 8×2 and at most 8×2 cap beam. The shallower companion reduces these configurable budgets to 32 reparent seeds, one additional layer, 24 component seeds, one round, and a 6×1 beam.

For clarity, the retained size boundaries are collected here:

mechanism	production boundary
complete labeled-tree frontier	$n \le 8$
primary intensive search	$n \le 32$
centered-disabled companion branch	$26 \le n \le 30$
centered fixed-MST DP	$n \le 128$
quality cap schedule	$n \le 256$
additional MST/BRBC families	$n \le 512$

(38)

The centered DP is evaluated at four caps

$$D = \Delta + H, \qquad H \in \{\lfloor 0.7\Delta \rfloor, \lfloor 0.75\Delta \rfloor, \lfloor 0.8\Delta \rfloor, \Delta\},$$

again replacing a zero positive-instance threshold by 1.

When $k < 64$, the scalable engine imports its cheaper endpoint schedule before filtering, and the quality engine imports the scalable schedule. Quality also imports the scalable schedule throughout the transition band $33 \le n \le 40$, even at $k = 64$. These unions protect lower-cost anchors when a small output cap changes which nondominated seeds survive; they do not make the empirical policies approximation algorithms.

C.5 General portfolio truncation

Suppose the exact nondominated frontier is

$$T_0, T_1, \ldots, T_{m-1}, \quad L(T_0) < \cdots < L(T_{m-1}), \quad R(T_0) > \cdots > R(T_{m-1}).$$

If $m \le k$, it is returned unchanged. Otherwise the empirical selector reserves the two endpoints. For each candidate marked essential by an anchor or lower-policy import, it also reserves a frontier point that dominates the marked candidate and minimizes exact additive slack. If these reservations themselves exceed k, endpoints are retained first and the remaining reserved indices are admitted in frontier order until the cap is reached. Thus unconditional witness preservation is a property of the separate certified selector, not of an arbitrarily small empirical portfolio.

For every consecutive pair T_a, T_b already selected, each unselected T_i, $a < i < b$, receives the exact triangle score

$$A(a, i, b) = |(L_b - L_a)(R_i - R_a) - (L_i - L_a)(R_b - R_a)|. \qquad (39)$$

The selector repeatedly inserts the point with largest A, breaking ties by frontier index, until k points have been selected. Equation (39) is a deterministic shape-preserving downsampling rule; it is not a hypervolume optimizer and carries no approximation claim.

C.6 Claim and formalization boundaries

The guarantees used in the paper have three distinct statuses.

- The weak NP-completeness result and the height-partition inequalities are mathematical theorems proved in this paper. The executable certificate checks are independent arithmetic audits of a produced witness, not machine-checked proofs of those theorems.
- The centered dynamic program is exact only for the fixed-MST family defined by (32)–(33). Complete labeled-tree enumeration via Prüfer codes is exact for unrestricted terminal trees only at the enforced cutoff $n \le 8$.
- Only CERTIFIED, on distinct signed-64-bit-coordinate terminals, has the universal $(2, 2)$ guarantee. Claims for FAST, BALANCED, and QUALITY are empirical claims about the released benchmark records.

The released Lean snapshot contains completed, `sorry`-free modules for the finite tree model and elementary lower bounds, Manhattan parity, and the gcd length lattice. It does not contain completed formal proofs of the NP-completeness reduction, the height-partition theorem, the metric lower bound, or Theorem C.2. Accordingly, none of those results is presented as formally verified.

D Experimental Protocol and Complete Frozen Results

This appendix fixes the meanings of every reported comparison and records the complete aggregate results in the release bundle. The machine-readable JSONL files, rather than the rounded values below, are authoritative.

D.1 Independent evaluator

An instance consists of distinct integer-coordinate terminals $V = (v_0, \ldots, v_{n-1})$, with v_0 as the root. A solver output is a parent array p, where $p_0 = -1$. Before using an output, the evaluator checks the parent range, rejects self loops and cycles, verifies that every chain reaches v_0, and then recomputes

$$L(p) = \sum_{i=1}^{n-1} \|v_i - v_{p_i}\|_1, \qquad R(p) = \max_i \sum_{(u,w)\in p[v_0, v_i]} \|u - w\|_1 \tag{D.1}$$

using Python integers. Values printed by the solver are never trusted. Duplicate parent arrays and duplicate (L, R) pairs are removed. Sorting the remaining pairs by increasing L, a pair is retained exactly when its R is strictly smaller than every earlier radius; this produces the exact nondominated frontier $\mathcal{F}(P)$.

For two portfolios P and Q, define

$$P \succeq Q \iff \forall q \in \mathcal{F}(Q)\ \exists p \in \mathcal{F}(P) : L(p) \le L(q) \ \wedge\ R(p) \le R(q). \tag{D.2}$$

The four reported relations are *dominance* ($P \succeq Q$, $Q \not\succeq P$), *tie* ($P \succeq Q$, $Q \succeq P$), *mixed* (neither), and *loss* ($Q \succeq P$, $P \not\succeq Q$). Thus "tie" means equality after Pareto closure, not equality of parent arrays or generator tags.

Hypervolume is also fixed per instance. If the largest coordinates in the frozen reference frontier are $L_{\max}$ and $R_{\max}$, the common corner is

$$C_L = L_{\max} + \max\{1, \lfloor L_{\max}/20 \rfloor\},$$
$$C_R = R_{\max} + \max\{1, \lfloor R_{\max}/20 \rfloor\}.$$

For frontier points (L_i, R_i) in increasing-L order, put $R_0 = C_R$ and compute

$$\mathrm{HV}_C(P) = \sum_{i:\, L_i < C_L,\ R_i < R_{i-1}} (C_L - L_i)(R_{i-1} - R_i).$$

The per-instance score is

$$\Delta\,\mathrm{HV}(P) = \frac{\mathrm{HV}_C(P) - \mathrm{HV}_C(P_{\mathrm{ref}})}{C_L C_R}.$$

All area calculations before the final division are exact integers. We report both the sum over 28 instances and its mean; these two aggregations must not be compared as though they were the same quantity.

Two different common factors. The empirical common factor of a whole portfolio is

$$c_{\mathrm{port}}(I, P) = \min_{T\in P} \max\left\{\frac{L(T)}{M_I}, \frac{R(T)}{\Delta_I}\right\}. \tag{1}$$

By contrast, the certificate factor is attached to the particular $H = \Delta_I$ height-partition tree C_I:

$$c_{\mathrm{cert}}(I) = \max\left\{\frac{L(C_I)}{M_I}, \frac{R(C_I)}{\Delta_I}\right\} \le 2. \tag{2}$$

Equation (1) may select the MST, star, or another height threshold, and is therefore normally smaller than (2). If C_I is Pareto dominated, the returned portfolio may retain an exactly remeasured dominator instead; the proof obligation is still checked against the reconstructed C_I. No benchmark in this appendix is degenerate ($M_I \Delta_I > 0$).

D.2 Datasets and frozen reference

The 28-instance development set contains 4–32 terminals: one smoke case, one SALT toy case, four extracted Superblue nets, uniform and two-cluster random families, two grids, and two near-line cases. It was visible during development and is used for regression and baseline comparison, not as a blind generalization set.

The separate 50-instance set contains 6–100 terminals from uniform, three-cluster, near-line, grid, comb, and radial families. The scaling set contains five geometries—uniform, four-cluster, near-line, comb, and radial—at each $n \in \{64, 128, 256, 512, 1024, 2048\}$, for 30 instances total. Finally, exact validation uses three six-terminal instances ($6^4 = 1296$ labeled trees each) and one eight-terminal instance ($8^6 = 262144$ labeled trees).

The 28-case published-method reference is the nondominated union of three sources:

(1) terminal-only star, RMST, PD, BRBC, and KRY outputs from the pinned SALT snapshot;
(2) the official PD-Rev/PD-II implementation at the 19 values $\alpha_1 = \alpha_2 = k/20$, $k = 1, \ldots, 19$, flipping distance two, with HVW/DAS disabled;
(3) the official 2023 MSPD/MSS implementation at the 11 settings $\alpha = j/10$, $j = 0, \ldots, 10$, and the 16 pairs $(\kappa, \lambda) \in \{1, \ldots, 4\}^2$, exported immediately before HVW/DAS Steinerization.

Legacy inputs are translated by a common per-instance offset before being passed to the author programs; Manhattan distances are unchanged. Every author output is converted to a terminal parent array and independently remeasured. After Pareto closure, the published union contains 139 stored points: 56 classic, 51 PD-II, and 32 MSPD/MSS points. Environment 04 is not part of this union; its outputs are used only in the separately labeled internal-predecessor stress test below.

D.3 Raw-data provenance

The frozen exporter ran all modes with a portfolio limit of 64 and a 120-second per-invocation timeout. The quality and held-out sets retain one run per case and mode; scaling retains three runs for each of 30 cases and four modes. There is no discarded warm-up. Wall time starts immediately before process creation and ends after process reap and output flush; per-process peak RSS is obtained from `wait4` when available. Thread environment variables are fixed to one, and the release build uses portable `-O3 -DNDEBUG` settings without architecture-specific flags.

The result environment records source commit `bb4e95c0e08`; the complete hash is retained in the machine-readable metadata. The raw bundle contains 112 development records, 200 held-out records, 360 scaling records, 78 certificate reconstructions, four exact-oracle records, and 168 ablation records. Each invocation record includes the command, complete input, stdout, stderr, exit status, timeout status, wall time, independently measured tree metrics, and frontier. Environment, hypervolume-corner, compiler, and verification metadata are stored separately. The CSV and Markdown summaries are generated only from these raw records.

D.4 Frozen quality results

Table 2 measures the entire frontier. In particular, the published union covers the certified portfolio on 26 instances, is mixed with it on one, and ties it on one; the factor-two theorem does not imply frontier coverage. Balanced and quality cover the union on every instance and add at least one strict Pareto improvement on 23.

D.5 Internal-predecessor stress test

As a separate diagnostic, we supplemented the published-method union with the frozen output of Environment 04, an earlier solver produced within this project. This deliberately stronger stress-test union is neither an external baseline nor a state-of-the-art claim, and it is not used for the main comparisons above.

On the development set, the certified portfolio contains one to six nondominated trees (median three); on the held-out set it contains one to seven (median four). The worst development portfolio factor, 1.53, occurs on `uniform_n30_s2`. Independent certificate reconstruction finds no failure among all 78 development and held-out cases. The returned portfolio contains the original $H = \Delta$ tree in 66 cases (23 development and 43 held-out) and an exact Pareto dominator in 12 (5 and 7, respectively).

D.6 Held-out ordering and absolute time

The large maxima for balanced and quality arise from their bounded intensive small-instance tiers; they are retained rather than hidden by a cutoff. Table 5 is a portfolio statement, whereas the first three numeric columns of Table 6 summarize one balanced point per instance.

D.7 Scaling

All 360 scaling invocations are valid and completed within the timeout. These are candidate-only measurements; the scaling data do not contain baseline runs.

D.8 Exactness and component ablation

Complete labeled-tree enumeration via Prüfer codes gives exact frontiers for all four small instances. Fast, balanced, and quality tie the oracle on all four; certified loses on all four, as expected for its deliberately sparse construction family. The production test suite has 26 unit and property tests, and the frozen verification record reports zero quality-gate regressions, zero certificate failures, and successful builds of both recent baseline adaptations.

The variants are compiled from the same source. “Construction only” removes heuristic search; “scalable tier only” removes the intensive small-case tier; the next two rows remove the named search stages; and “reduced starts” lowers deterministic search trials and seed caps. Because constructions also seed later neighborhoods, these rows are interventions, not additive attributions.

D.9 Released-artifact boundary

The public repository releases the solver source and tests; the three benchmark JSONL files; the independent evaluator and result exporters; the frozen reference parent arrays; redistributable source, licenses, and provenance for the PD-II and MSPD/MSS adaptations; selected 01–05 prompts and task contexts; the completed, `sorry`-free Lean Basic, Parity, and Gcd modules; and the frozen raw outputs and generated summaries described above.

It does *not* redistribute the pinned SALT source: only SALT’s validated terminal-tree contribution to the frozen union is present. It also does not claim to release an exhaustive agent transcript or hidden model state. Incomplete Lean developments for NP-completeness, height partition, and the approximation lower bound are omitted, so the main theorems are not claimed as formally verified. These boundaries are part of the reproducibility statement, not exceptions to it.

Table 2: All four modes against the 28-case published-method union.

Mode	Dominance	Tie	Mixed	Loss	$\sum_I \Delta\mathrm{HV}$	$\mathrm{mean}_I\ \Delta\mathrm{HV}$
certified	0	1	1	26	−0.7478	−0.026708
fast	12	6	9	1	+0.0654	+0.002337
balanced	23	5	0	0	+0.0863	+0.003081
quality	23	5	0	0	+0.0863	+0.003081

Table 3: Diagnostic comparison with the published union augmented by the internal Environment-04 predecessor.

Mode	Dominance	Tie	Mixed	Loss	$\sum_I \Delta\mathrm{HV}$
certified	0	1	0	27	−0.6038
fast	0	11	5	12	−0.0127
balanced	13	15	0	0	+0.0050
quality	13	15	0	0	+0.0050

Table 4: Common-factor statistics. "Certified portfolio" uses Eq. (1); "$H = \Delta$ certificate" uses Eq. (2).

Set	Object	Mean	Median	Min.	Max.
development	certified portfolio	1.21	1.24	1.00	1.53
development	published portfolio	1.06	1.05	1.00	1.18
development	$H = \Delta$ certificate	1.27	1.25	1.00	1.94
held-out	certified portfolio	1.25	1.22	1.00	1.97
held-out	$H = \Delta$ certificate	1.31	1.25	1.00	1.97

Table 5: Coverage transitions on all 50 held-out instances. A transition $A \to B$ compares candidate B with reference A.

Transition	Dominance	Tie	Mixed	Loss
certified → fast	44	3	3	0
fast → balanced	21	29	0	0
balanced → quality	9	41	0	0

Table 6: Held-out common factor and one-process wall time. Times include startup and serialization; one run is retained per case.

Mode	Mean c_{port}	Median c_{port}	Maximum c_{port}	Median time (ms)	Maximum time (ms)
certified	1.25	1.22	1.97	6.9	13.7
fast	1.11	1.08	1.44	13.0	262.0
balanced	1.11	1.08	1.44	62.2	19030.5
quality	1.11	1.08	1.44	63.3	19140.5

Table 7: Median end-to-end wall time in milliseconds on the scaling set. Each entry is the median of five geometries and three retained runs.

n	certified	fast	balanced	quality
64	6.1	8.7	8.7	9.8
128	6.3	17.0	16.9	19.9
256	6.1	15.3	15.6	26.2
512	9.4	18.1	18.5	22.7
1024	13.3	42.2	41.9	54.2
2048	27.1	140.8	142.2	182.1

Table 8: Complete $n = 2048$ endpoint. Peak RSS is the maximum observed per-process resident set; tree count is the median.

Mode	Median ms	Max. ms	Trees	Peak MiB
certified	27.1	32.5	5	2.48
fast	140.8	182.8	8	3.66
balanced	142.2	220.4	8	3.75
quality	182.1	302.3	8	4.05

Table 9: Complete compact ablation against the 28-case published-method union. Runtime columns are from the same ablation run and are included to expose its variable host load; quality comparisons do not use them.

Variant	Dom.	Tie	Mix.	Loss	$\sum \Delta$HV	Median ms	Max. ms
production	23	5	0	0	+0.0863	848.8	37991.6
construction only	1	9	8	10	−0.0589	16.5	602.3
scalable tier only	12	7	9	0	+0.0665	72.0	1008.9
no deep search	20	6	2	0	+0.0840	670.2	5981.2
no component exchange	23	5	0	0	+0.0857	706.4	30436.9
reduced starts	23	5	0	0	+0.0862	785.7	38378.8

E Released Research Record and Formalization Boundary

This appendix states exactly which parts of the model-assisted research record are public. It distinguishes a *task specification*, which records what an investigation was allowed to see and asked to do, from an execution transcript, which would record everything that happened while carrying out that task. The repository releases the former for five conditions, but not the latter.

E.1 Model setting and five task specifications

All five investigations used OpenAI GPT-5.6 Sol in Codex with ultra reasoning effort. They differed in their assigned objective, information boundary, and minimum persistence requirement, not in the selected model or reasoning setting.

Each directory under `prompts/` contains exactly three files: `PROMPT.md`, `AGENTS.md`, and `PROBLEM.md`. The first two record the intended objective and access policy; the third gives the common problem definition. At a high level, the five conditions were:

- `01`: **hardness, closed world.** The condition supplied the problem definition but no references, algorithms, benchmarks, or Internet access. It asked for an adversarially audited NP-completeness proof, including the complete-Manhattan-graph shortcut audit, and required contemporaneous claim, approach, and counterexample ledgers.
- `02`: **exact tractability, blind.** This condition fixed the premise that a polynomial-time constructive route exists, prohibited hardness arguments and early abandonment, and supplied no papers or existing algorithms. It allowed a public benchmark and a command-line evaluator only as falsification aids, forbade inspecting or modifying the evaluator, and imposed an eight-active-hour persistence protocol.
- `03`: **exact tractability, locally informed.** The mathematical premise and persistence requirement matched `02`, but this condition could consult a curated local `resources/` collection and supplied baseline code. Network access and sibling experiments remained forbidden, and every transferred claim or code path required a provenance and model-match audit. The contrast with `02` therefore isolates the effect of prior literature and code without changing the requested complexity conclusion.
- `04`: **approximation, literature open.** This condition took the hardness result from `01` as a fixed premise and redirected effort toward a strong bicriteria approximation and an improved empirical frontier. Focused Internet search for public mathematical papers and implementations was permitted and source-logged; the frozen evaluator and supplied baselines remained immutable. The prompt required at least eight active hours and continued post-pass work on the theorem, algorithm, runtime, and held-out behavior.
- `05`: **full-context engineering synthesis.** The final condition could inspect the accumulated evidence from `01–04`, the local reference and baseline collections, and relevant public online material. It was asked to build and audit the strongest practical multi-mode solver, with exact output validation, controlled comparisons, ablations, and scaling measurements. Its four-active-hour budget was a minimum rather than a deadline, and the prompt deliberately left the final algorithmic architecture open.

These descriptions concern the *specified* information boundaries. The released prompt directories do not include the original `worklog/` or `submission/` trees, evaluator-private state, downloaded reference collections, integrity manifests, or filesystem images mentioned by those specifications. In particular, the presence of detailed ledger requirements inside a prompt must not be read as a claim that the corresponding complete ledger has been released. The public packages document the experimental conditions; they are neither full conversation histories nor containerized replays of the five runs.

E.2 What the public repository contains

The released repository contains the final deterministic solver and its build file; production, property, exact-small, and certificate tests; the checked-in quality, held-out, and scaling instances; an independent tree/frontier checker and result exporters; and the source and provenance material needed for the released PD-II and 2023 MSPD/MSS terminal-output adaptations. It also contains the frozen nondominated baseline union, algorithm and evaluation notes, the five task specifications above, the completed Lean snapshot described below, and machine-readable raw results together with tables generated from those raw records.

It does *not* release complete model conversations, complete condition-by-condition research histories, every intermediate claim ledger or counterexample ledger, every exploratory program, private reference copies, unfinished Lean developments, or complete execution-environment images. Likewise, the frozen recent-baseline artifact has no timing data collected under the solver's process-level timing protocol; it supports objective-space comparisons, not a same-machine runtime ratio. These omissions are why the artifact is described as a reproducibility subset of the research record rather than as "everything produced" during the project.

E.3 Lean snapshot and theorem boundary

The directory `formal/` contains three completed, `sorry`-free modules:

- `Rcrst/Basic.lean` defines finite metrics and rooted trees in the parent/depth model, total length, root radius, and direct radius. It proves the pointwise direct-distance lower bound, $\Delta \leq R(T)$, and that the star attains $R = \Delta$; it also defines the MST optimality predicate.
- `Rcrst/Parity.lean` constructs the integer Manhattan metric and proves that every root-to-terminal tree-path length has the same parity as the corresponding direct Manhattan distance.
- `Rcrst/Gcd.lean` proves the length-lattice lemma: if g divides every pairwise distance, then g divides every tree length, with the associated exact budget-rounding lemma.

The weak NP-completeness reduction in Appendix A, the height-partition theorem in Appendix B, and the associated lower-bound construction are *not* completed in Lean and are not included as Lean theorem files. The main results of this paper are consequently mathematical proofs checked in the ordinary manner, not claims of end-to-end formal verification. The Lean snapshot is also intentionally outside the default solver build and experimental verification target.

E.4 Frozen measurements and verification record

The `results/raw/` bundle stores per-instance machine-readable outputs for the 28-case comparison, 50 held-out cases, scaling runs through $n = 2048$, certificate reconstruction, exact-small checks, and component ablations. Invocation records retain the command, input, standard output and error, exit status, and wall time; scaling records additionally retain per-process peak resident memory when the host exposes it. The published CSV and Markdown summaries are regenerated only from these raw records.

The frozen verification record separately captures the portable optimized build, the 26-test suite, the public quality gate, and compilation of the two redistributable recent-baseline source trees. Its environment record includes the source revision, compiler and flags, operating-system and processor metadata, single-thread settings, timeout, timing boundary, and repeat policy. Lean is explicitly outside that verification target, consistently with the formalization boundary above. Together these files make the reported measurements inspectable and rerunnable without implying that an unreleased research transcript has been reconstructed.